

Assay for Transposase-Accessible Chromatin – sequencing (ATAC-seq)

Epigenomics Data Analysis Workshop

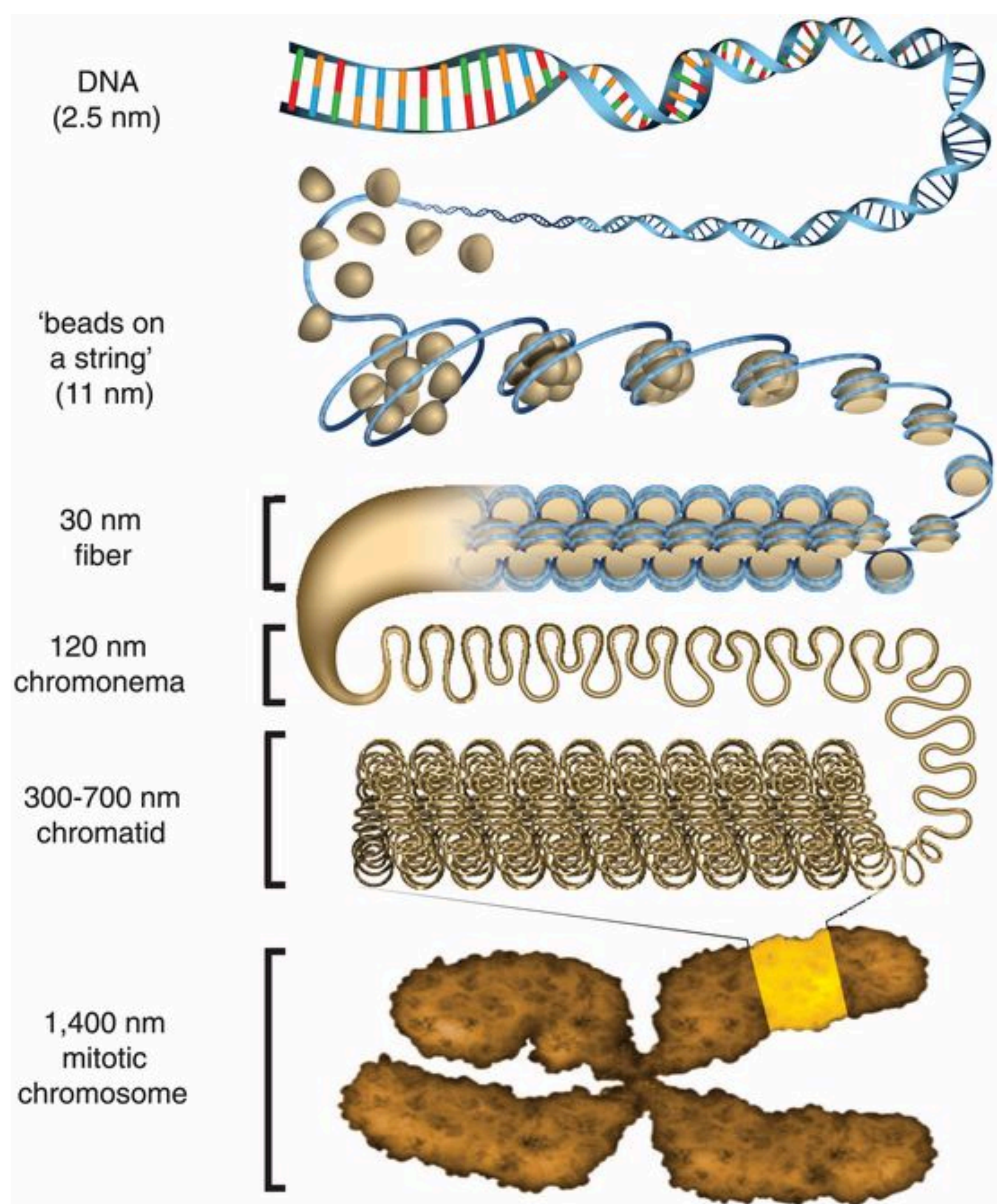
Stockholm, 6 October 2026

Agata Smialowska

NBIS, SciLifeLab, Stockholm University



Chromatin structure



Chromatin structure

ChIP-seq

ATAC-seq, DNaseI-seq,
MNase-seq

3D organisation
Hi-C, Dam-ID

DNA
(2.5 nm)

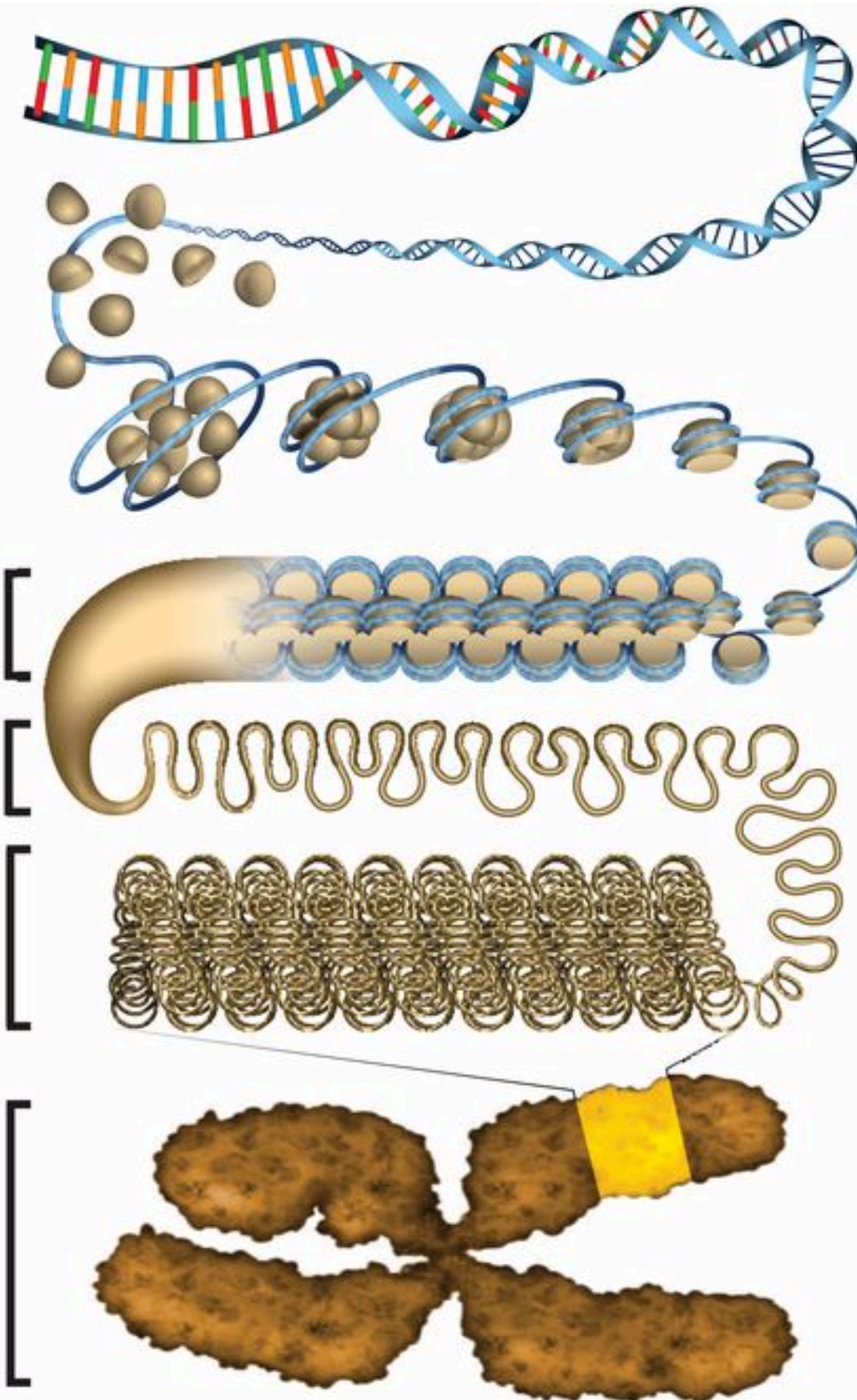
'beads on a string'
(11 nm)

30 nm
fiber

120 nm
chromonema

300-700 nm
chromatid

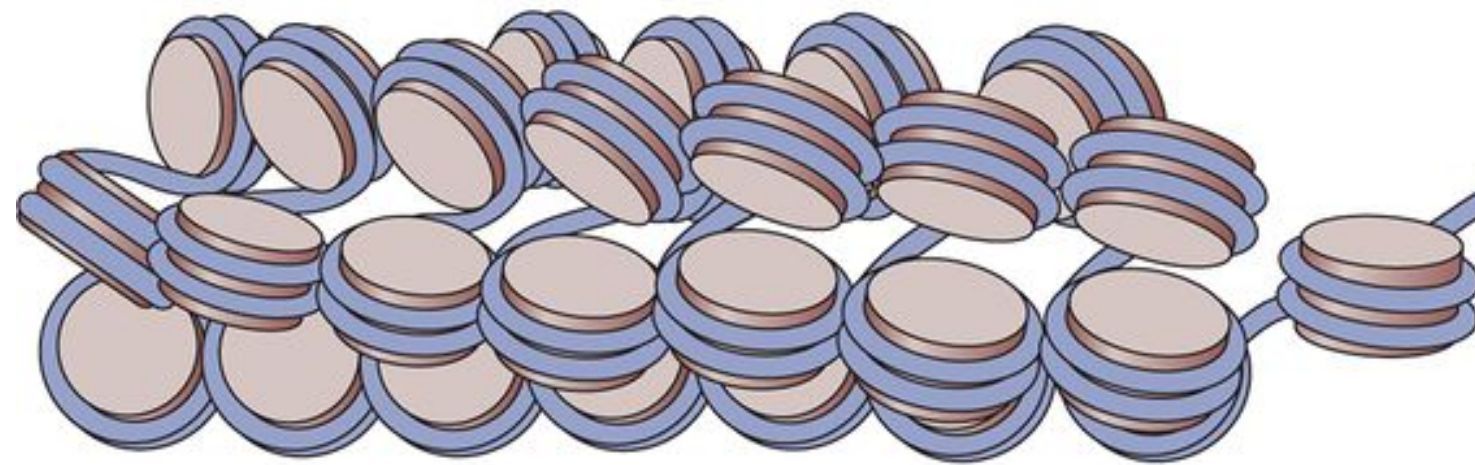
1,400 nm
mitotic
chromosome



Chromatin states and gene expression

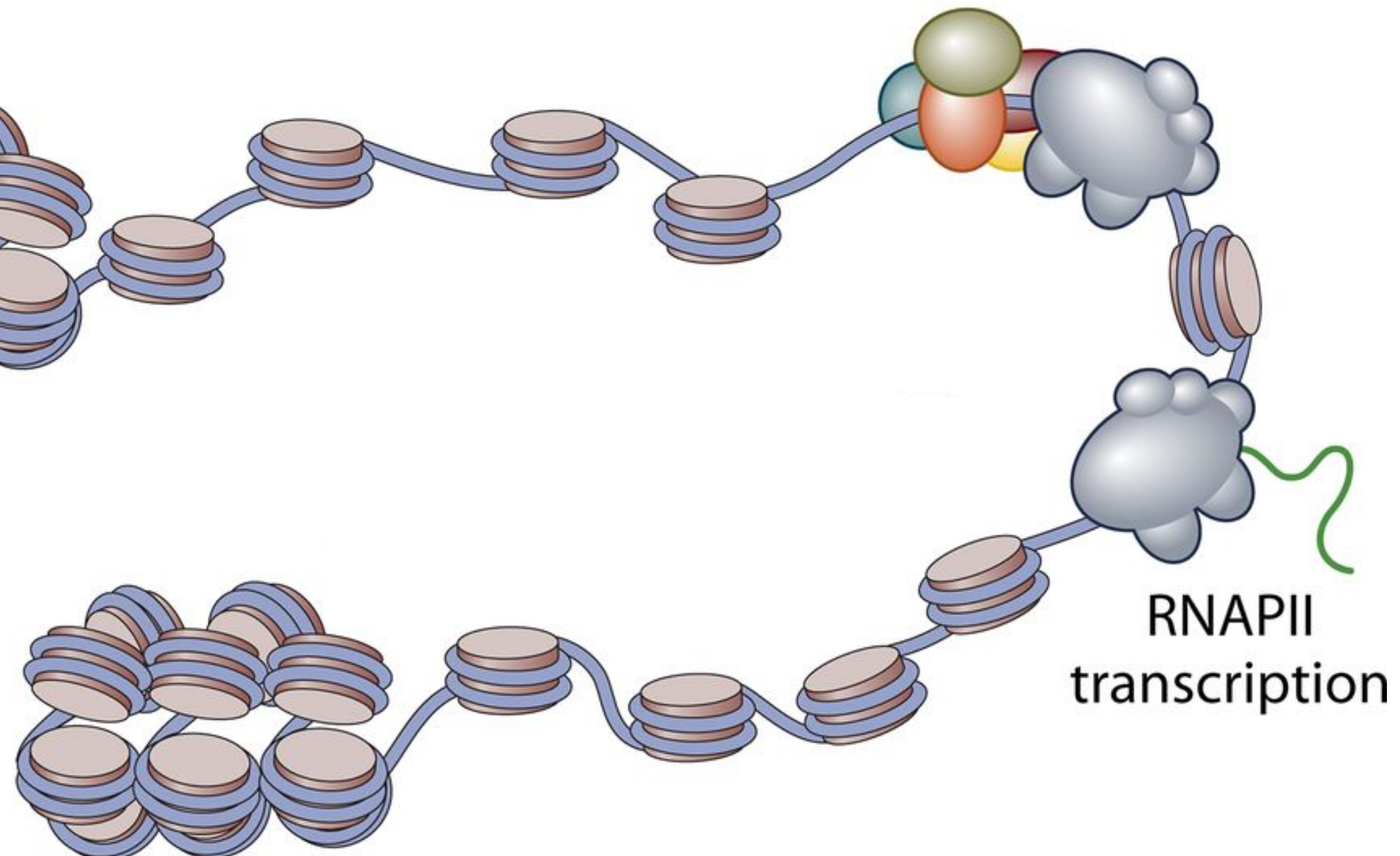
Gene suppression

"High" nucleosome density
"High" repressive methylation load
Hypoacetylation



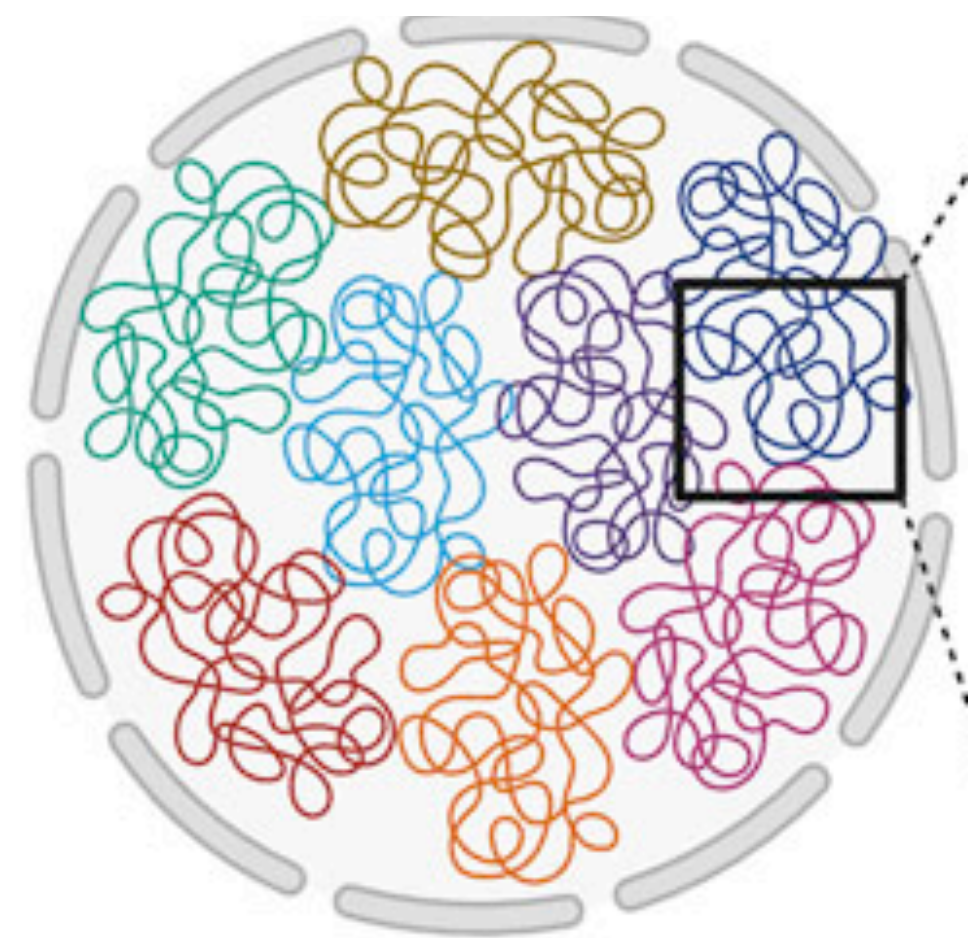
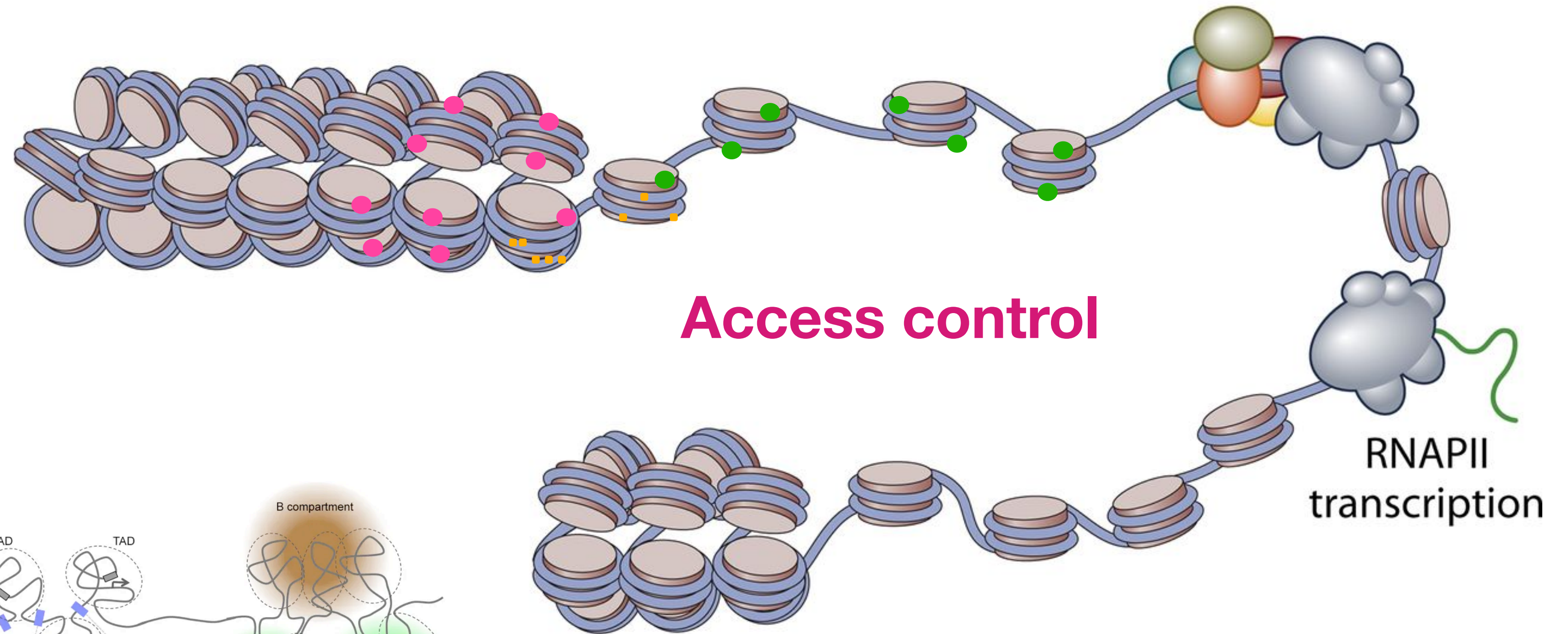
Gene activation

"Reduced" nucleosome density
Decreased repressive methylation load
Hyperacetylation

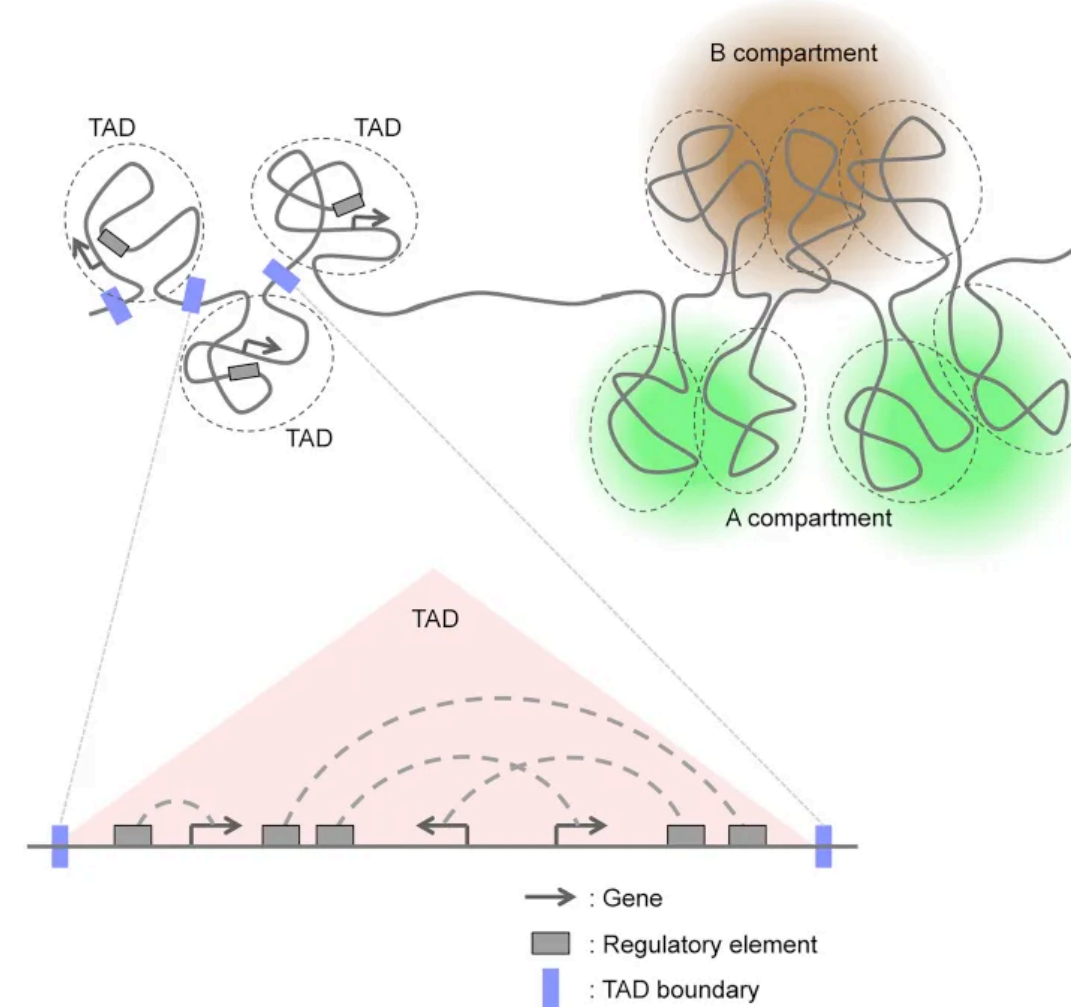


Chromatin states and gene expression

CpG-me
H3K27me3
H3K27ac

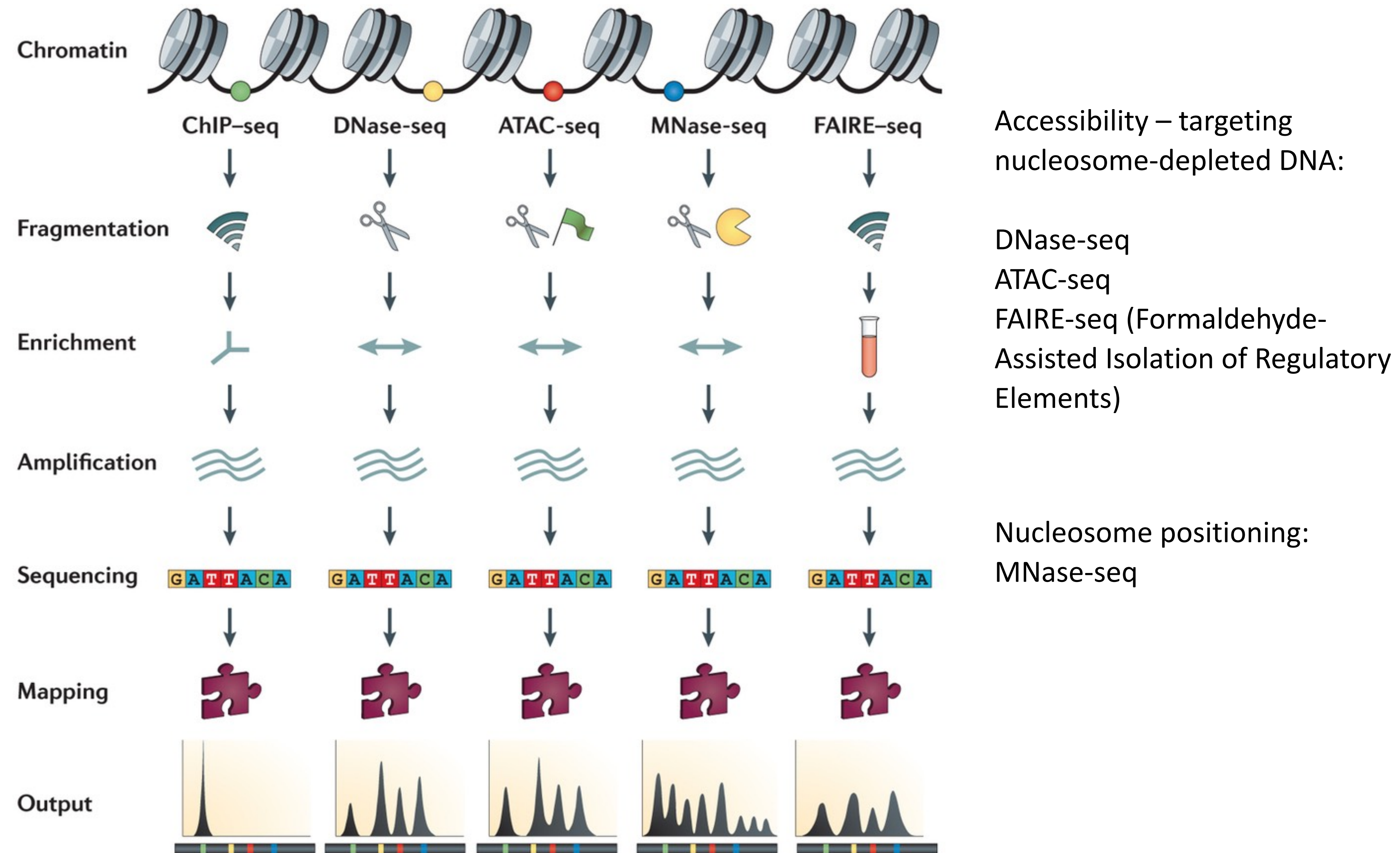


Chromosome Territories

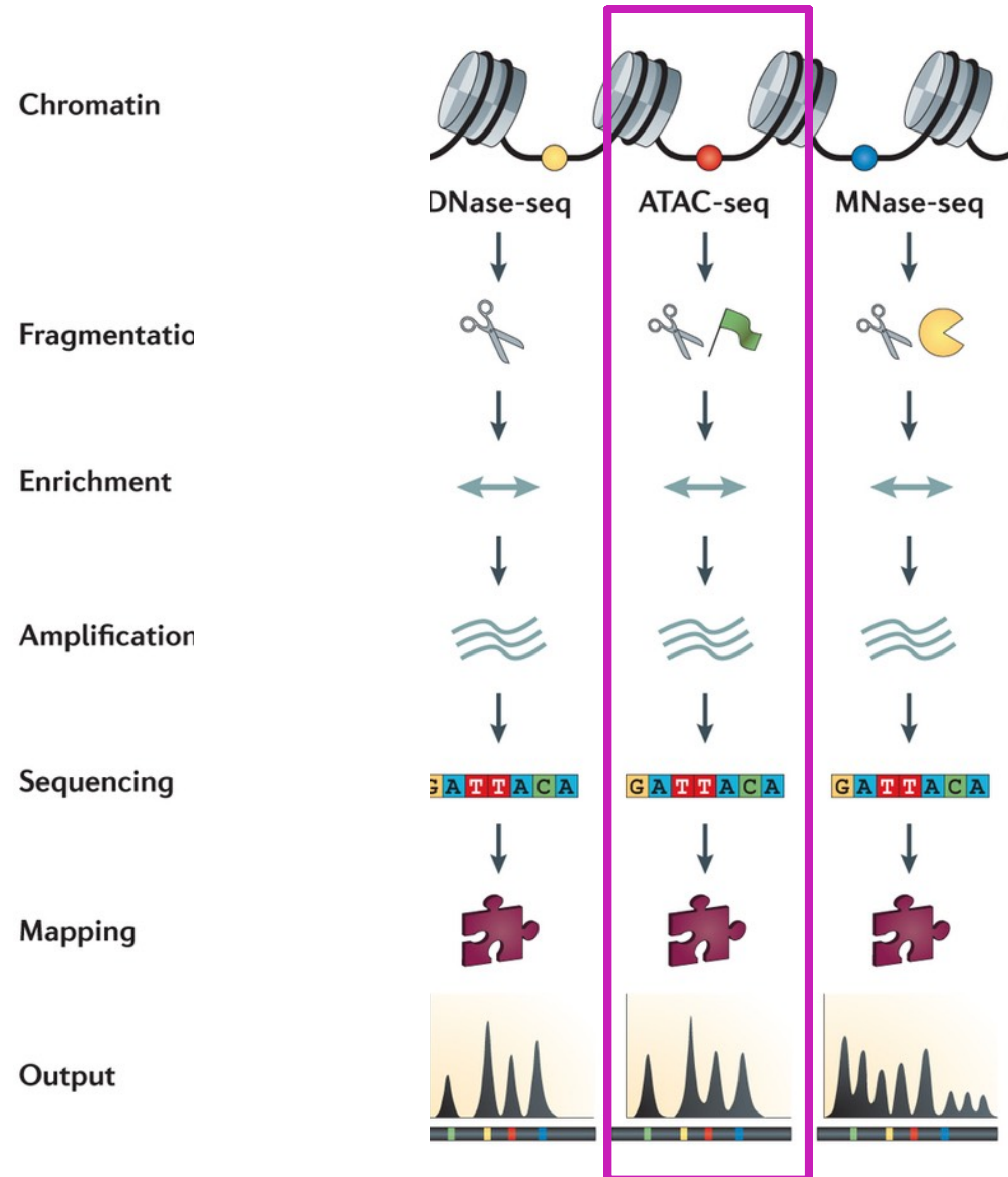


images: <https://doi.org/10.1128/mBio.00098-16>
Giles et al, 2025
<https://doi.org/10.3389/fnins.2018.00874>

Functional genomics techniques to probe chromatin states



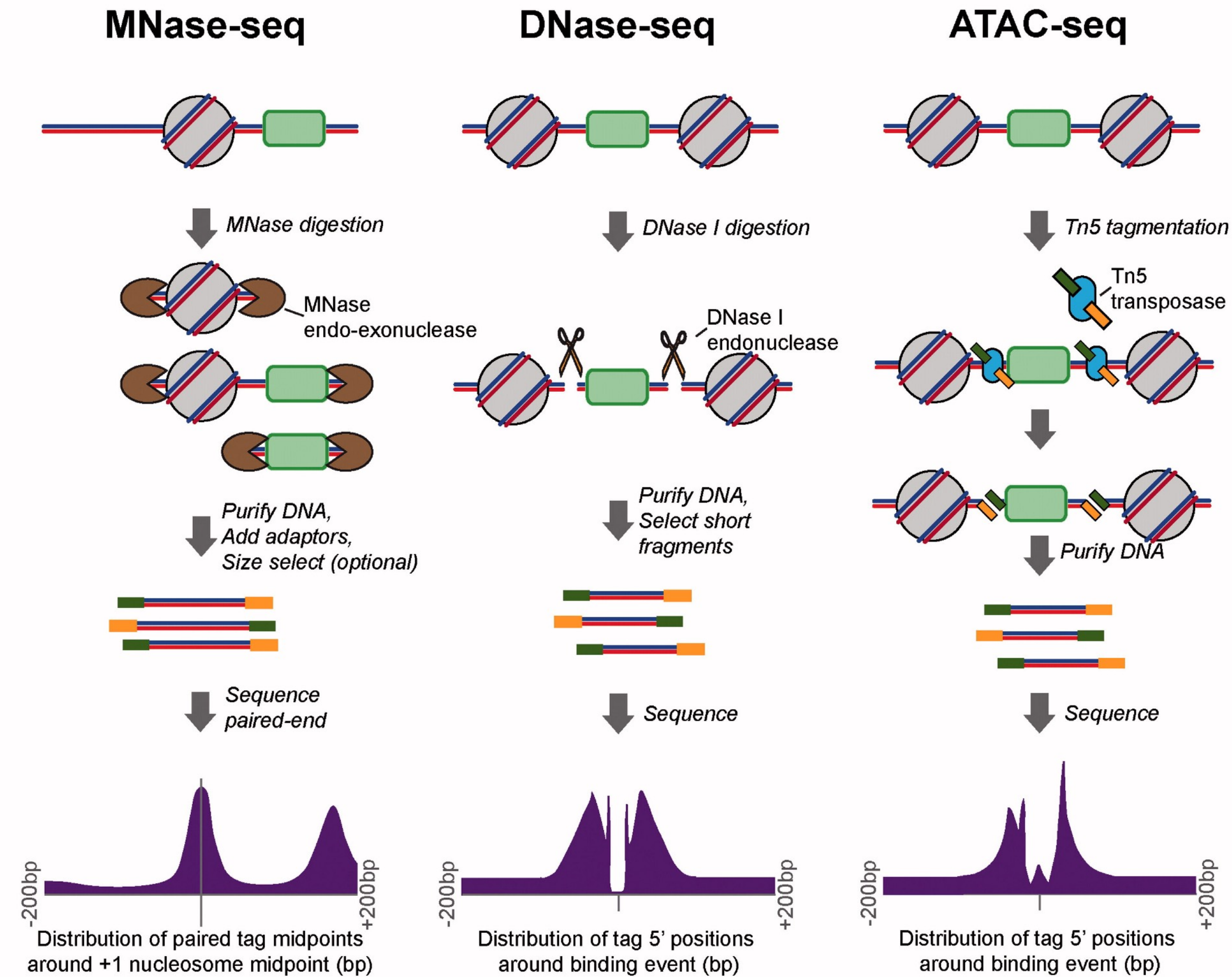
Functional genomics techniques to probe chromatin accessibility



Accessibility – targeting
nucleosome-depleted DNA:

ATAC-seq

Functional genomics techniques to identify open chromatin regions



Assay for Transposase-Accessible Chromatin (ATAC)-seq

- The method published in bulk (Buenrostro et al., 2015) and single cell (Buenrostro et al., 2015)

Current Protocols in Molecular Biology / Volume 109, Issue 1

UNIT

ATAC-seq: A Method for Assaying Chromatin Accessibility Genome-Wide

Jason D. Buenrostro, Beijing Wu, Howard Y. Chang, William J. Greenleaf

First published: 05 January 2015

<https://doi.org/10.1002/0471142727.mb2129s109>

Citations: 696

Published: 17 June 2015

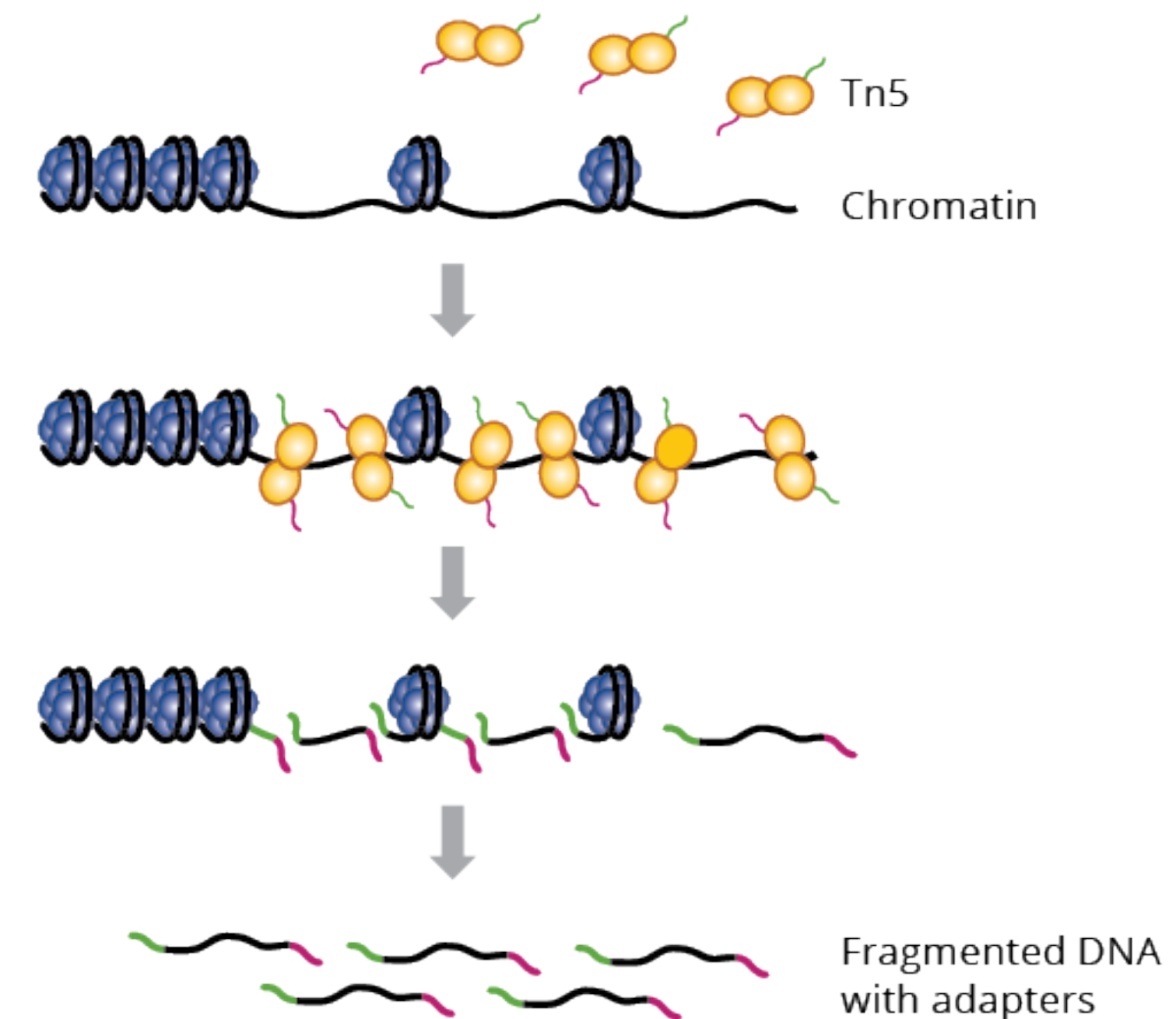
Single-cell chromatin accessibility reveals principles of regulatory variation

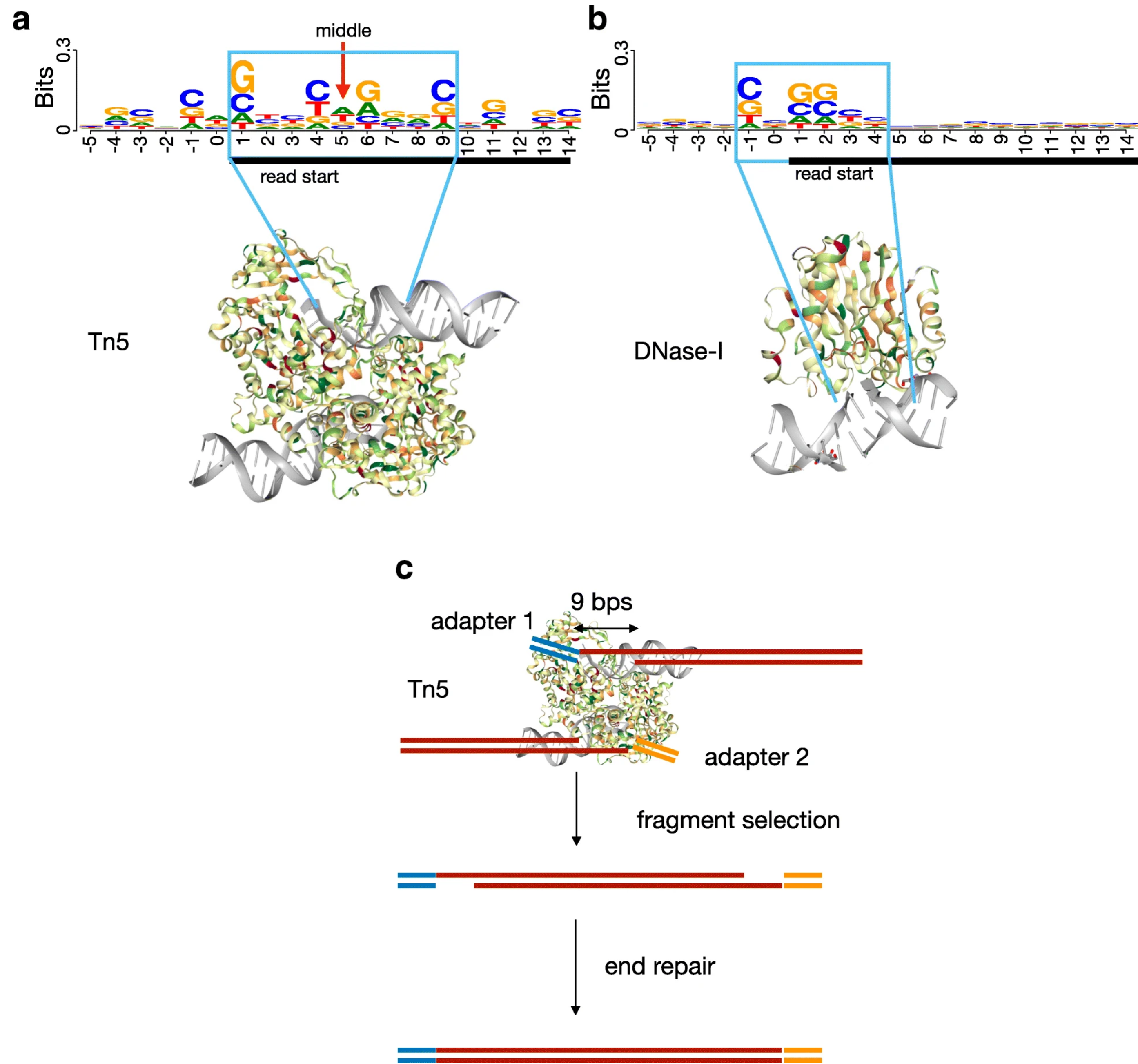
Jason D. Buenrostro, Beijing Wu, Ulrike M. Litzénburger, Dave Ruff, Michael L. Gonzales, Michael P. Snyder, Howard Y. Chang  & William J. Greenleaf 

Nature 523, 486–490(2015) | [Cite this article](#)

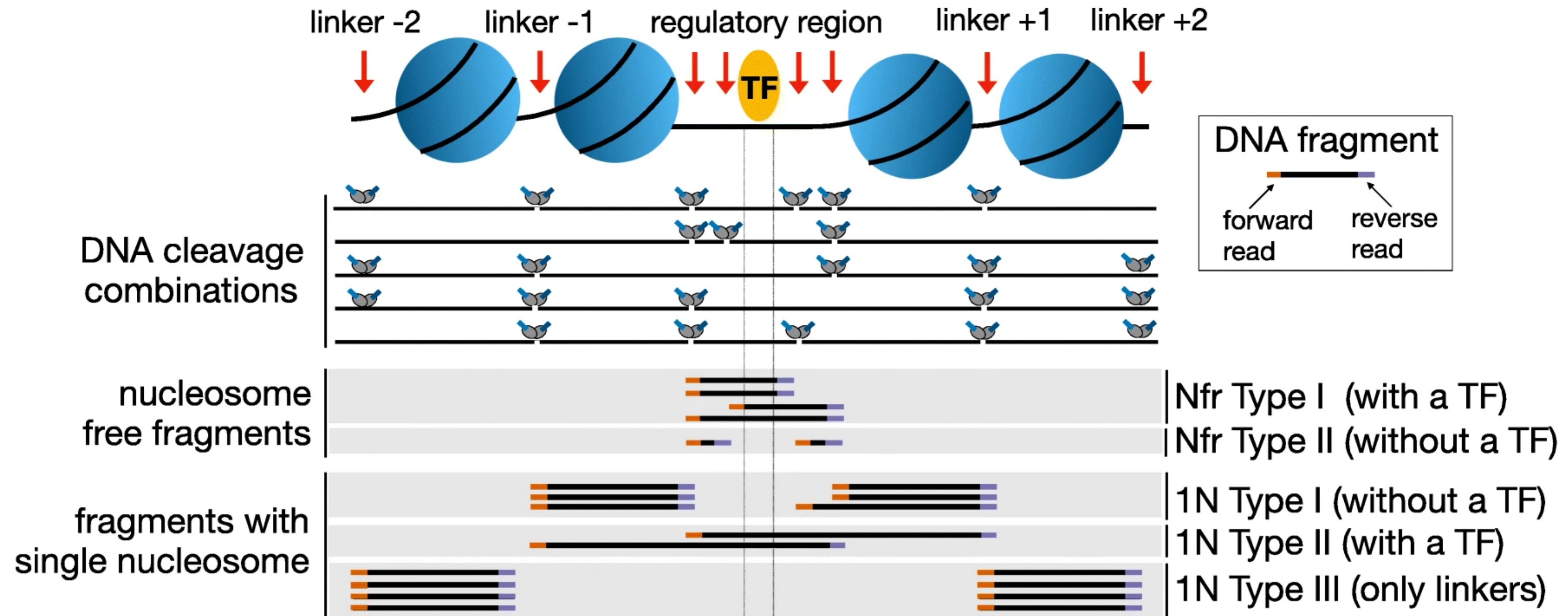
21k Accesses | 600 Citations | 100 Altmetric | [Metrics](#)

- It probes access to chromatin by using Tn5 transposase to insert sequencing adapters into DNA which allows simultaneous fragmentation of chromatin and integration of the adapters into open chromatin regions
- Significantly fewer cells needed (~ 50,000 cells for ATAC-seq compared to millions of cells for the other methods (DNase-seq or FAIRE-seq))
- Two step process, one day of work
- Applications: **accessibility**, nucleosome positioning at transcription start sites, transcription factor footprinting

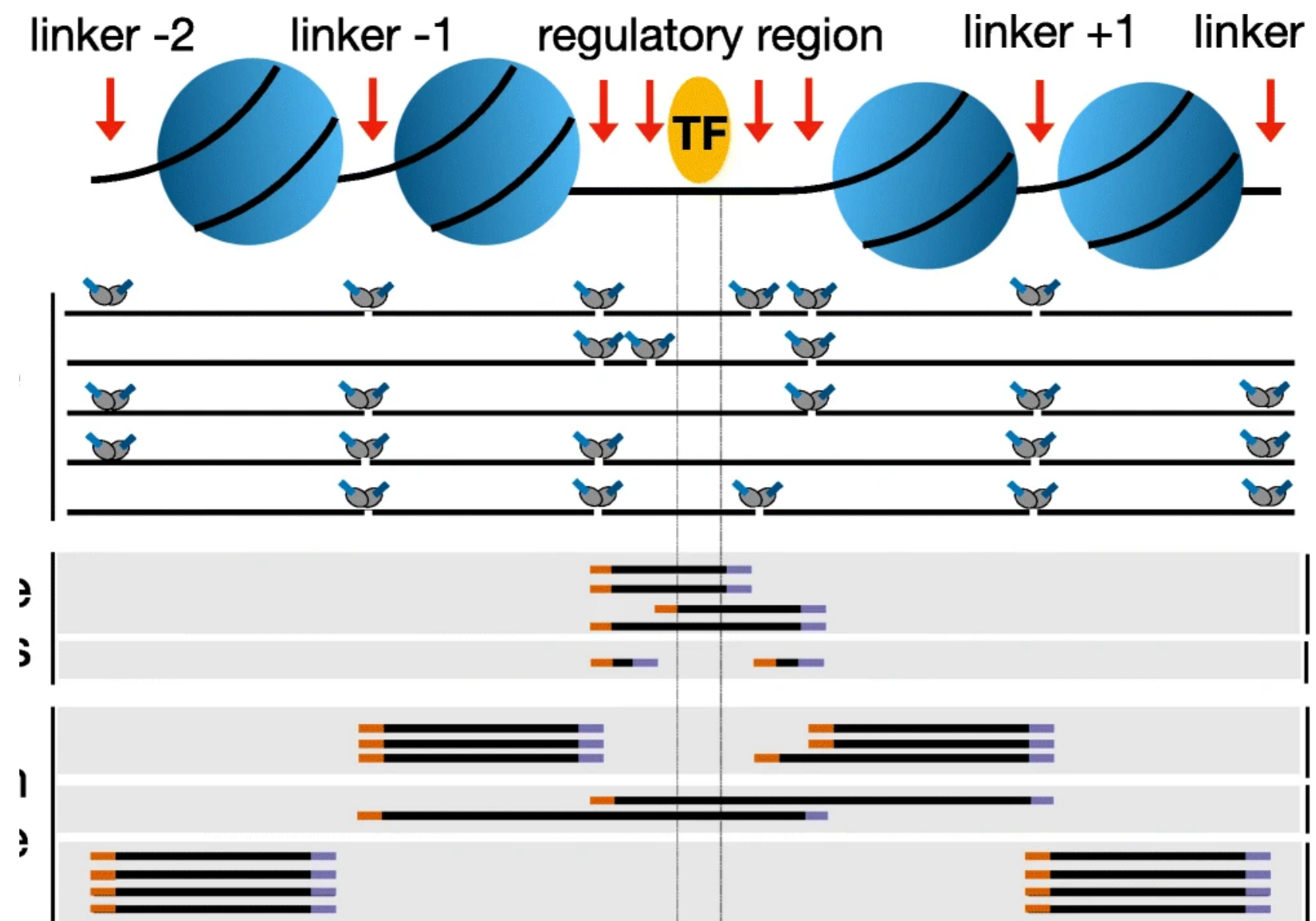




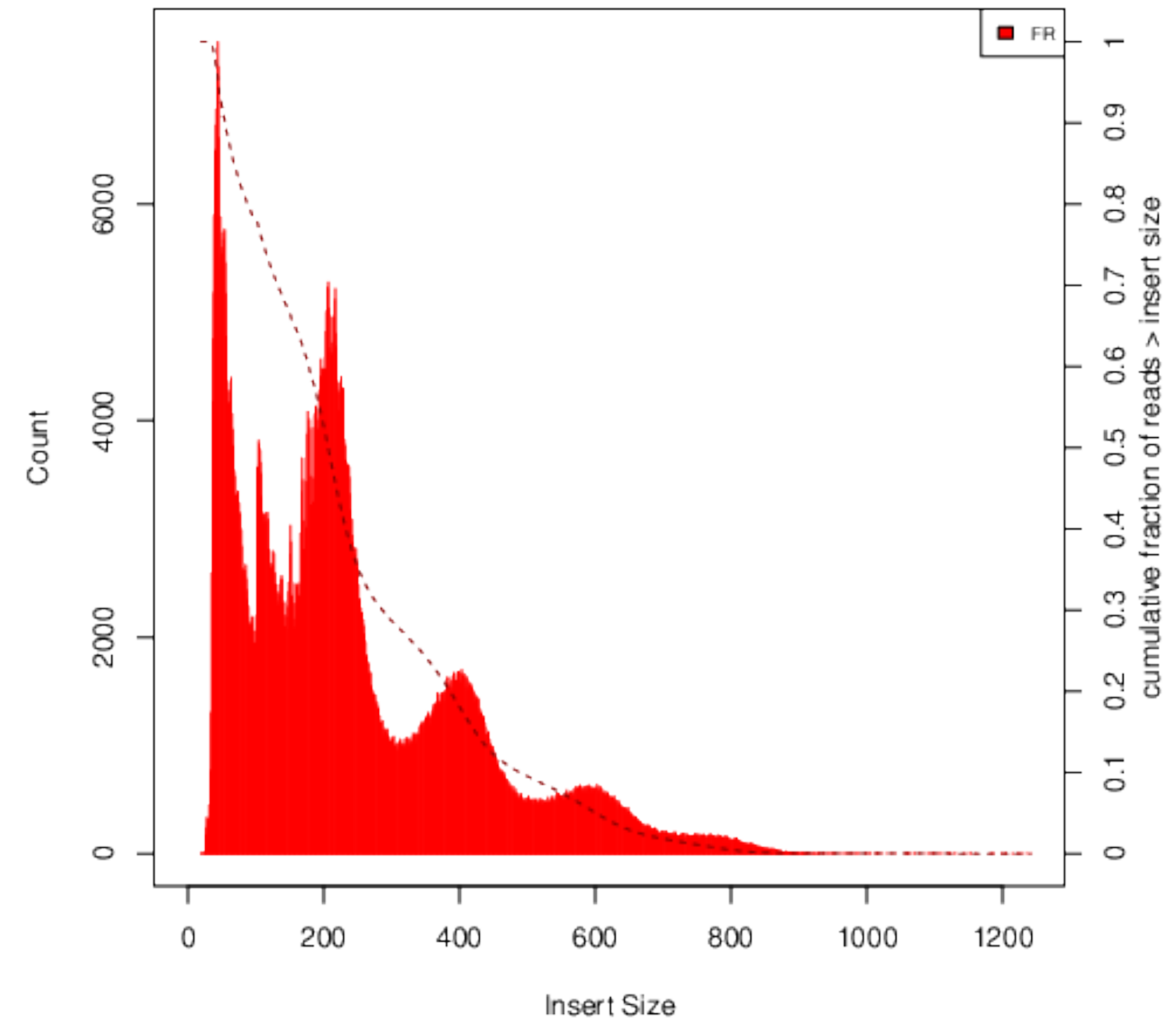
DNA fragments generated in ATAC-seq



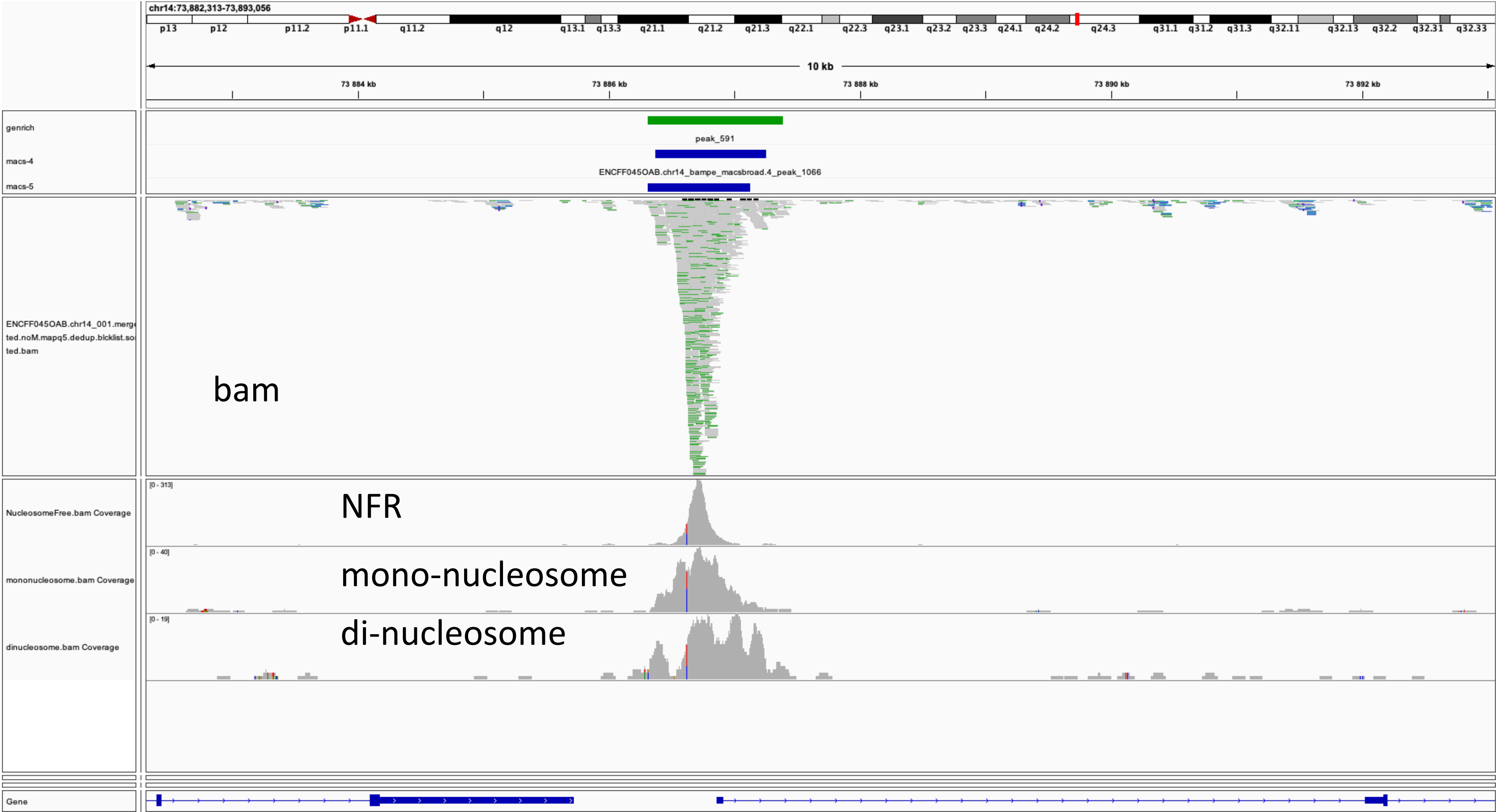
DNA fragments generated in ATAC-seq: QC



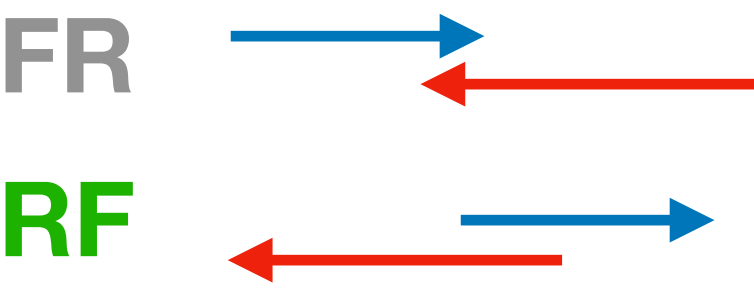
Insert Size Histogram for All_Reads
in file ENCFF045OAB.chr14.blacklist_M_filt.mapq5.dedup.bam



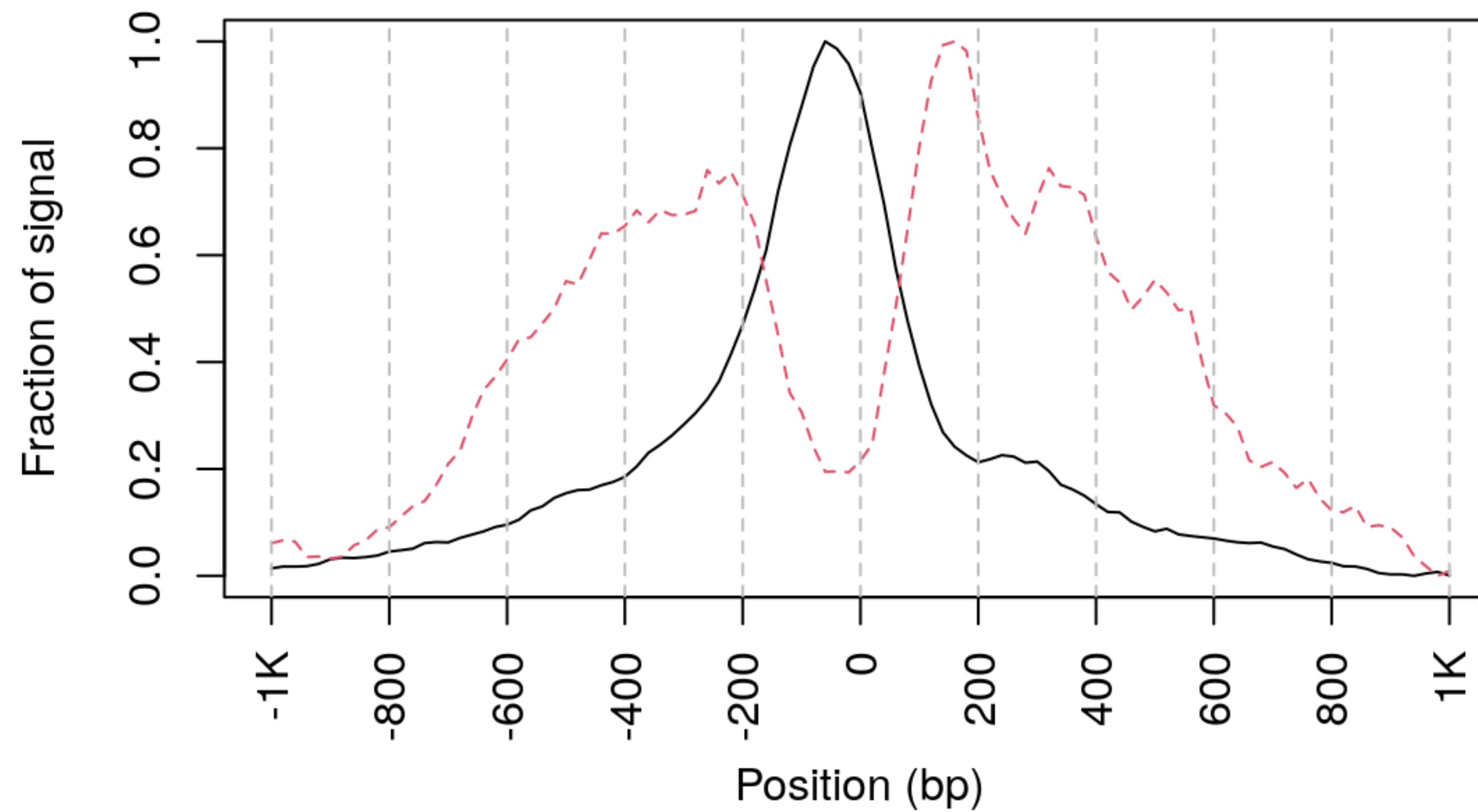
Distribution of nucleosome-free and mono nucleosome signal at TSS



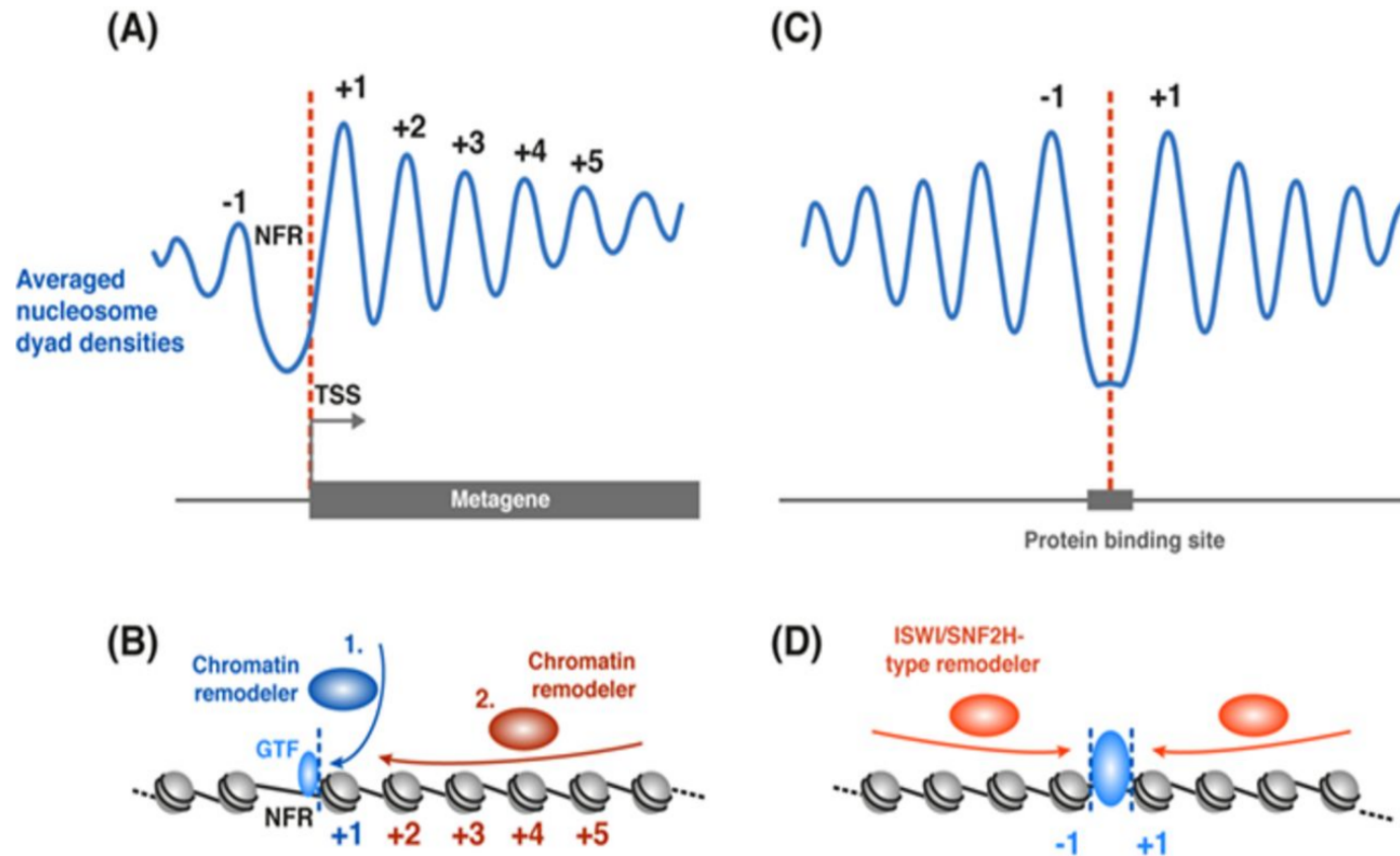
coloured by
fragment orientation



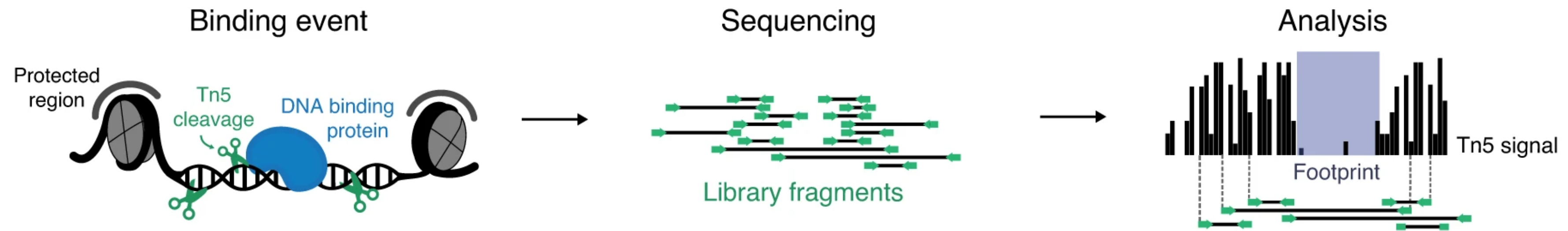
Distribution of nucleosome-free and mono nucleosome signal at TSS



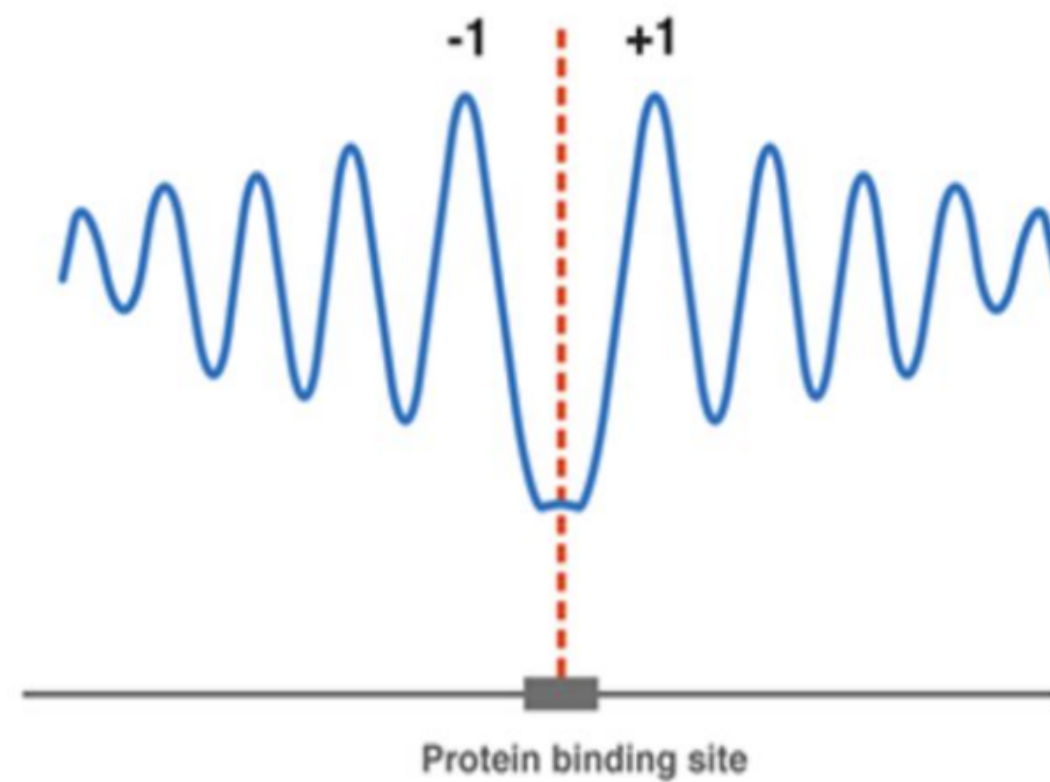
ATAC-seq: nucleosome resolution



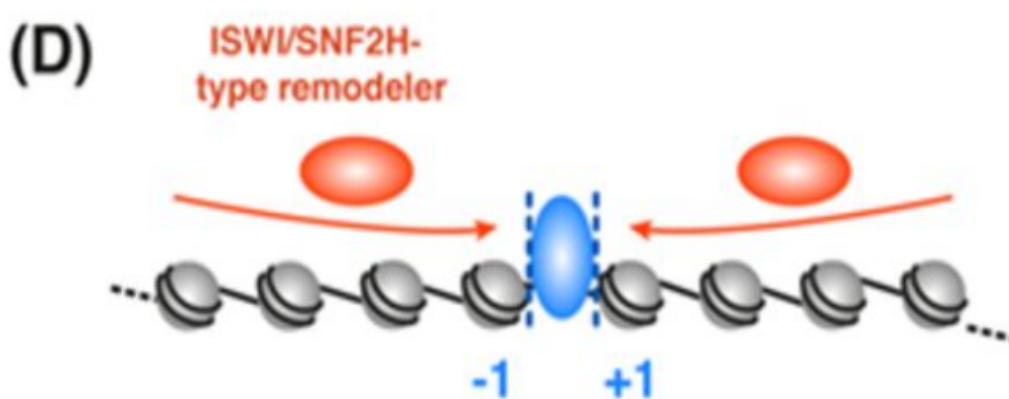
Transcription Factor Footprinting: Principle



(C)



(D)

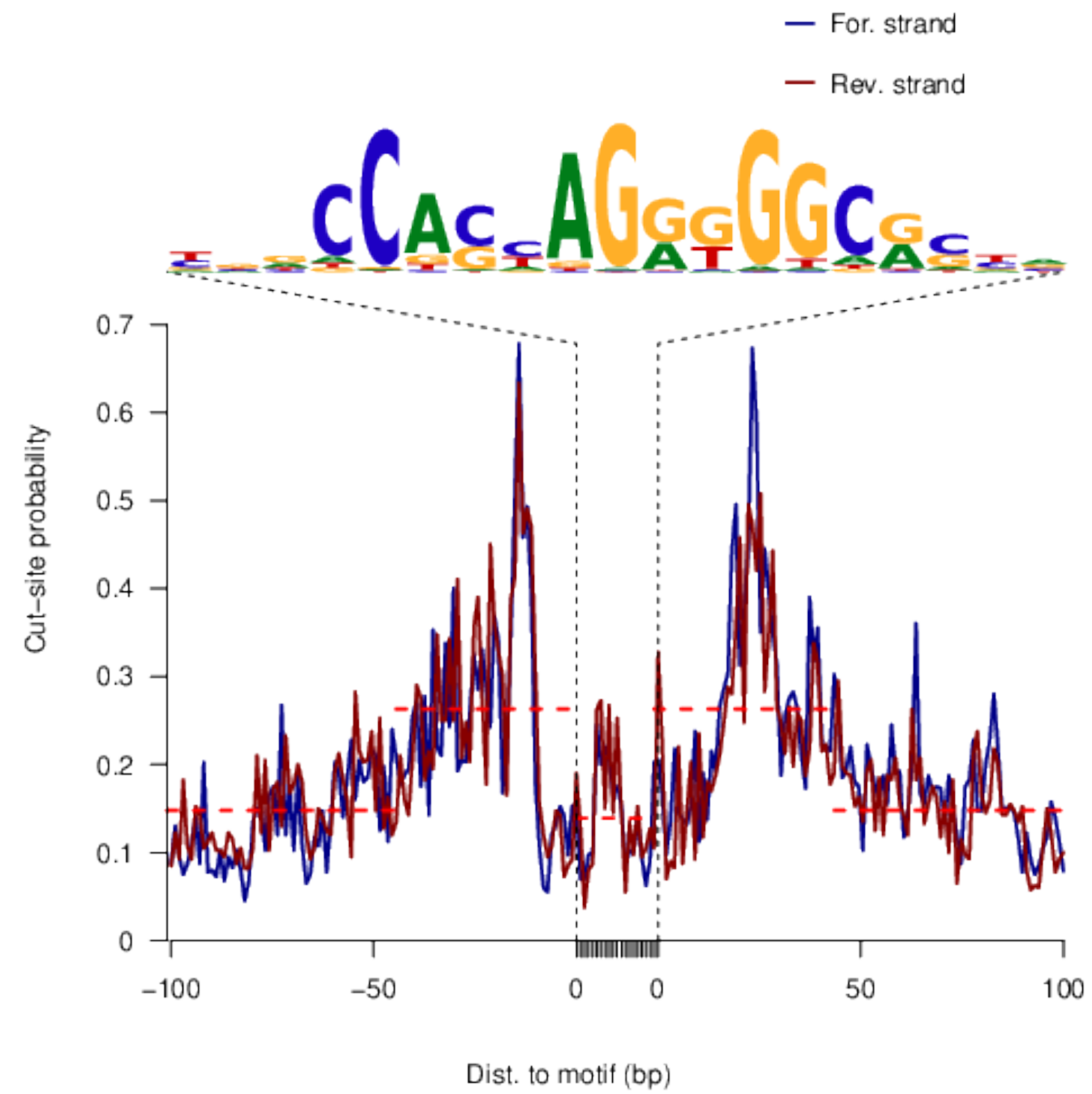


TF footprint

image:DOI:10.1038/s41467-020-18035-1

image: <https://doi-org.ezp.sub.su.se/10.1042/EBC20180058>

TF Footprinting - (a very simple) example (CTCF)



ATAC-seq data analysis

- Overview of data analysis steps
- Special considerations in experiment planning and quality metrics
- Differential accessibility in ATAC-seq
 - Normalisation
- Exercise overview
 - Data processing
 - Quality control
 - Peak detection
 - TF footprinting (tutorial planned for Thursday)

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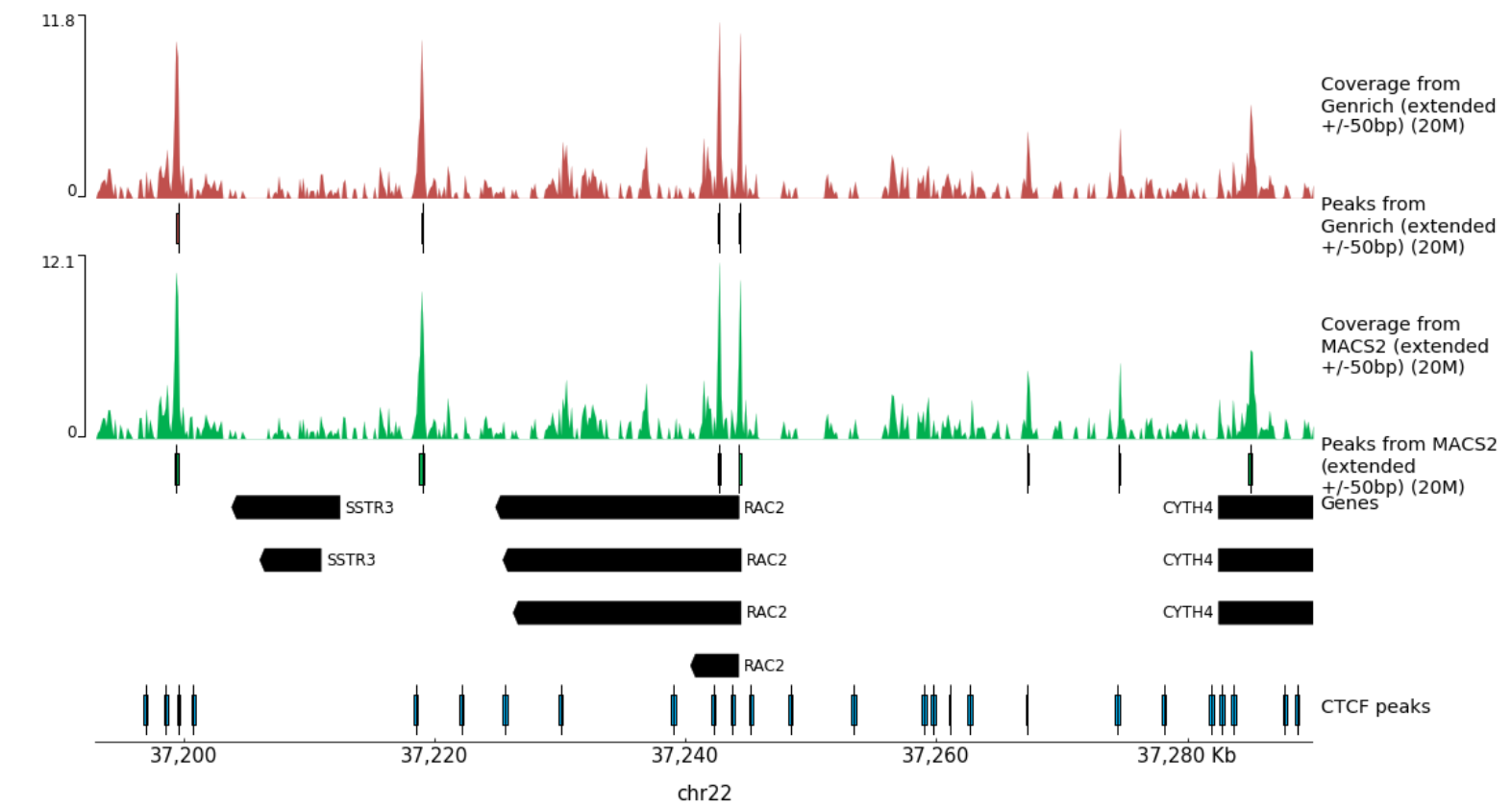
Analysis workflow for ATAC-seq data

raw data *.fastq



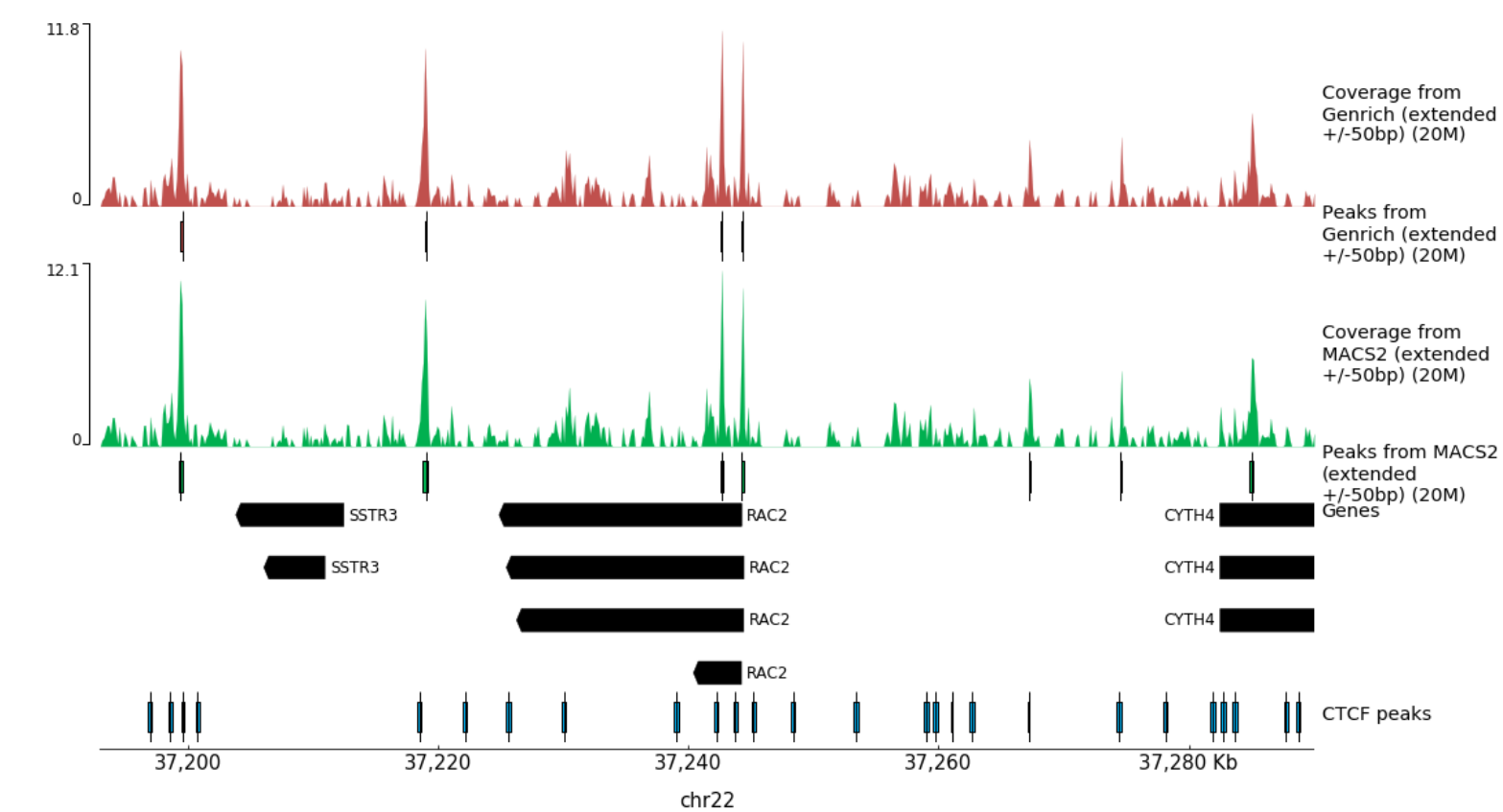
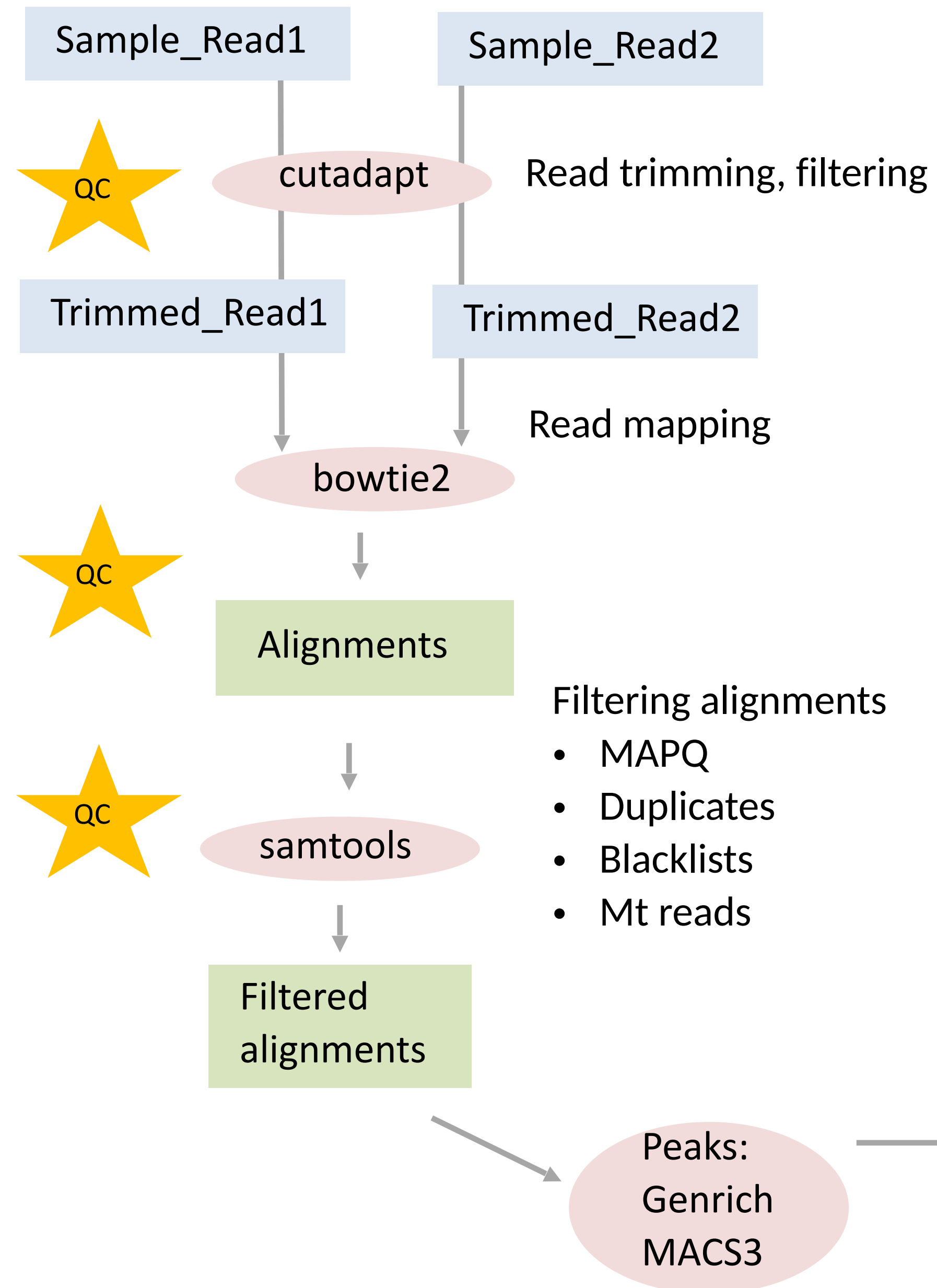
peaks & more

```
1 @ERR000589.41 EAS139_45:5:1:2:111/1
2 CTTTCCTCCCTGCTTTCCTGGCCCCACCATTTCAGGGAACATCTTGTCAT
3 +
4 3IIIIIIIIII>1IIIF9BG08E00I%IG+&?(4)%00646.C1#&(
5 @ERR000589.42 EAS139_45:5:1:2:1293/1
6 AGTTGTTAAAATCCAAGCCAATTAAGATAGTCTTATCTTTTTAAAAGAAAT
7 +
8 IIIIIIGII.AIIII=?I9G-/II=+I=4?761BA2C9I+5A711+&>1$/I
```

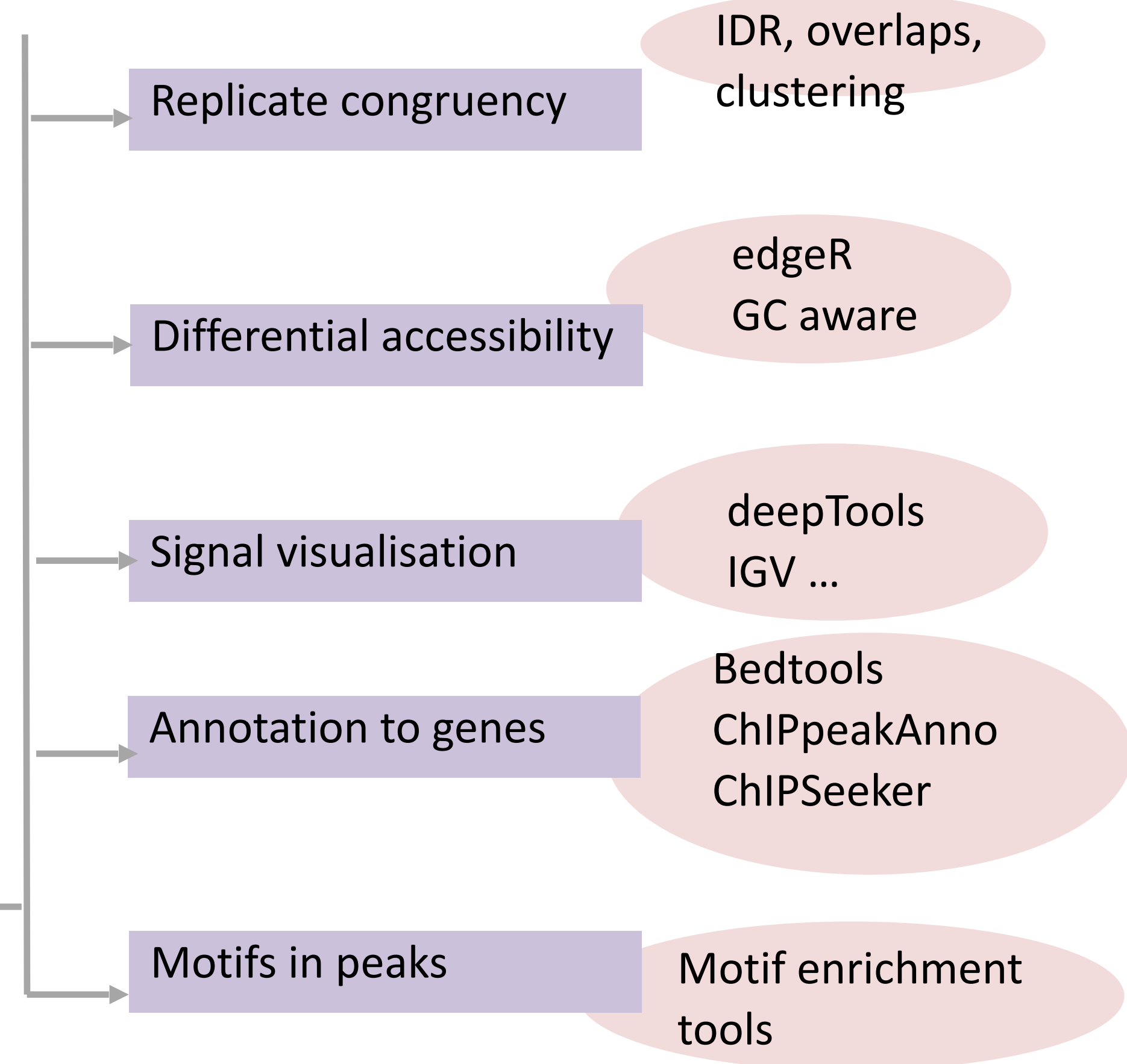


- fastq processing: quality and adapter trimming
- read mapping
- alignment filtering
- peak calling
- differential accessibility
- peak annotation
- visualisation
- TF footprinting
- TF motif analysis

Analysis workflow



...



Special considerations for ATAC-seq experiments

- Paired end (PE) sequencing is required
- QC: fraction of Mt reads – it can be high (up to 40%) – calculate sequencing depth accordingly
- QC: fragment length distribution – mononucleosome peak should be evident; consistent amongst libraries to be compared
- For current data quality standards, refer to ENCODE; currently:
 - 25 million non-duplicate, non-mitochondrial aligned read pairs (i.e. 50M PE reads);
 - non-redundant fraction >0.9;
 - fraction of reads in called peak regions (FRiP) >0.3;
 - mononucleosome peak present;
 - TSS enrichment observed

Special considerations for ATAC-seq differential accessibility analysis: effect of normalisation

Methodology | [Open Access](#) | [Published: 22 April 2020](#)

ATAC-seq normalization method can significantly affect differential accessibility analysis and interpretation

[Jake J. Reske](#), [Mike R. Wilson](#) & [Ronald L. Chandler](#) 

Epigenetics & Chromatin **13**, Article number: 22 (2020) | [Cite this article](#)

doi:<https://doi.org/10.1186/s13072-020-00342-y>

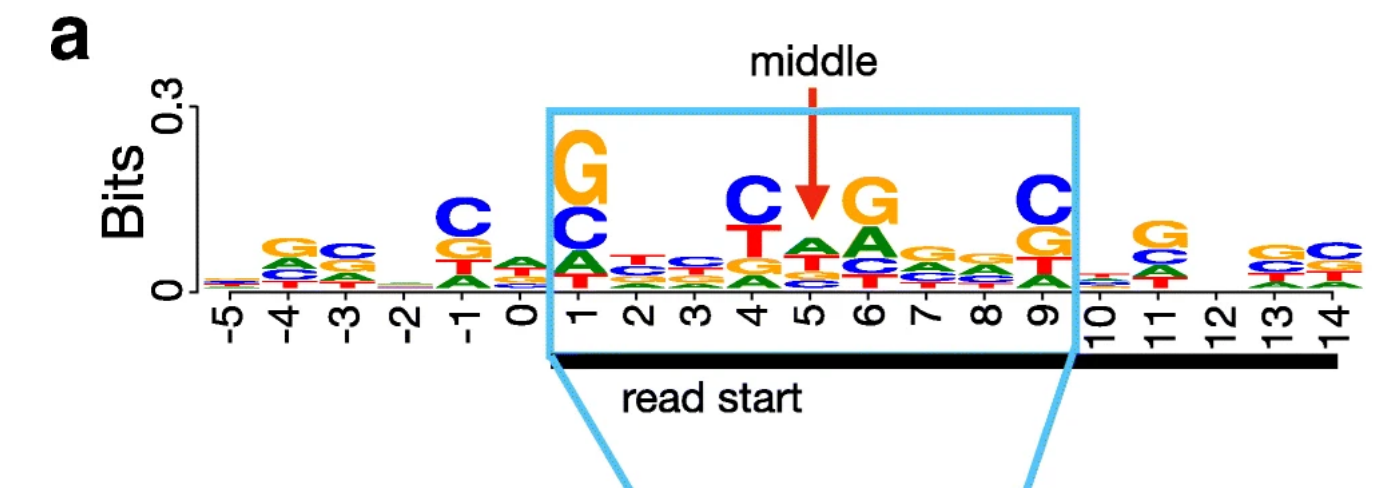
Normalization benchmark of ATAC-seq datasets shows the importance of accounting for GC-content effects

 Koen Van den Berge, Hsin-Jung Chou,  Hector Roux de Bézieux,  Kelly Street,  Davide Risso,
 John Ngai,  Sandrine Dudoit

doi: <https://doi.org/10.1101/2021.01.26.428252>

This article is a preprint and has not been certified by peer review [what does this mean?].

doi: <https://doi.org/10.1101/2021.01.26.428252>

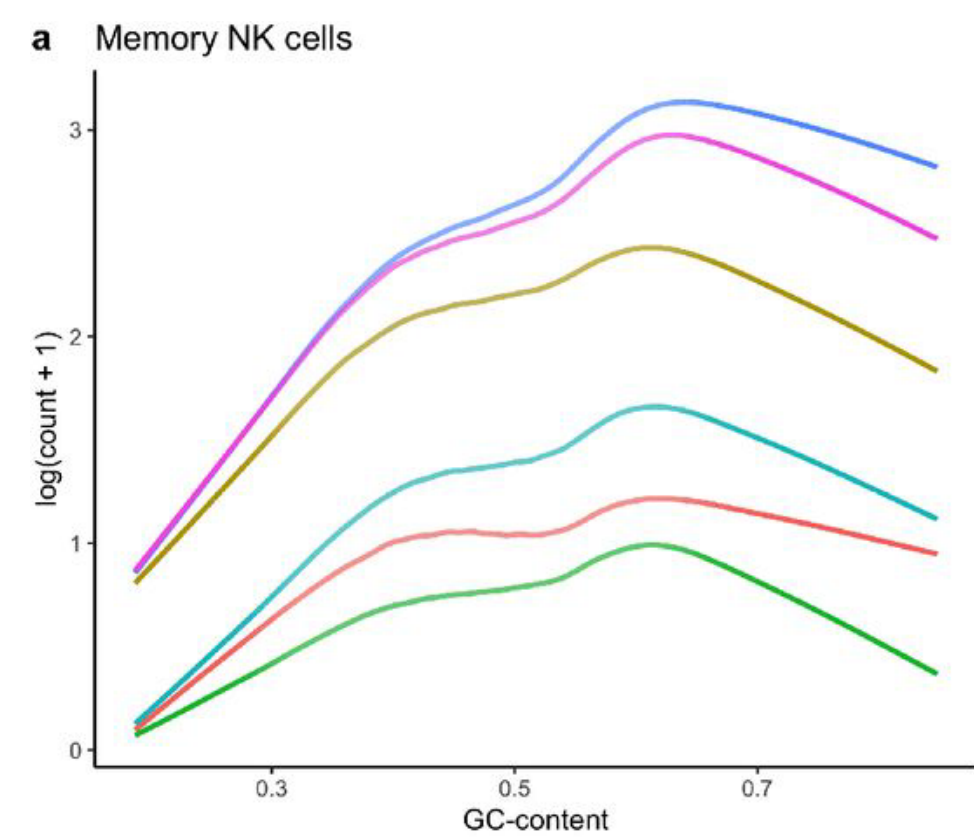


sequence bias in Tn5 target sites

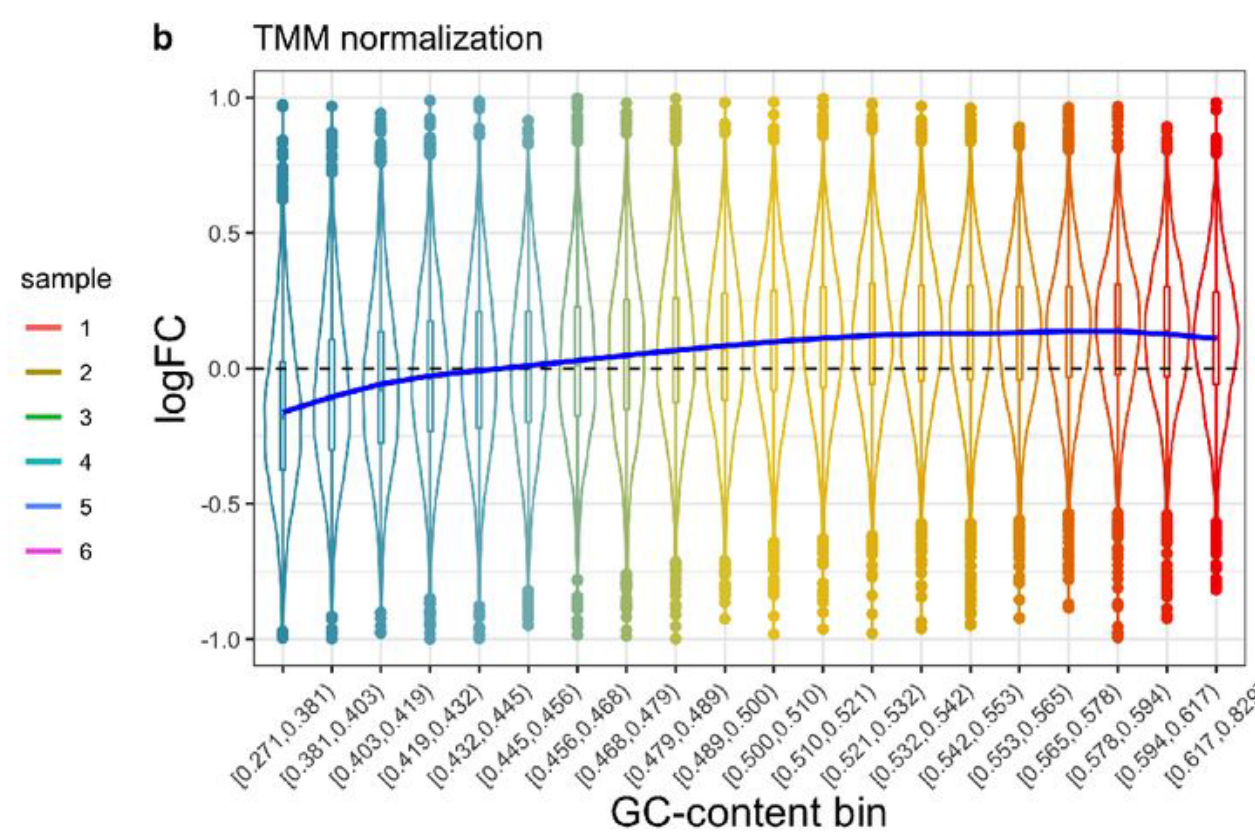
- GC-content effects are present in ATAC-seq datasets;
- Since the GC-content effects are sample-specific, they may bias downstream analyses such as clustering and differential accessibility;
- GC aware scaling normalisation methods exist;

Special considerations for ATAC-seq differential accessibility analysis: effect of normalisation

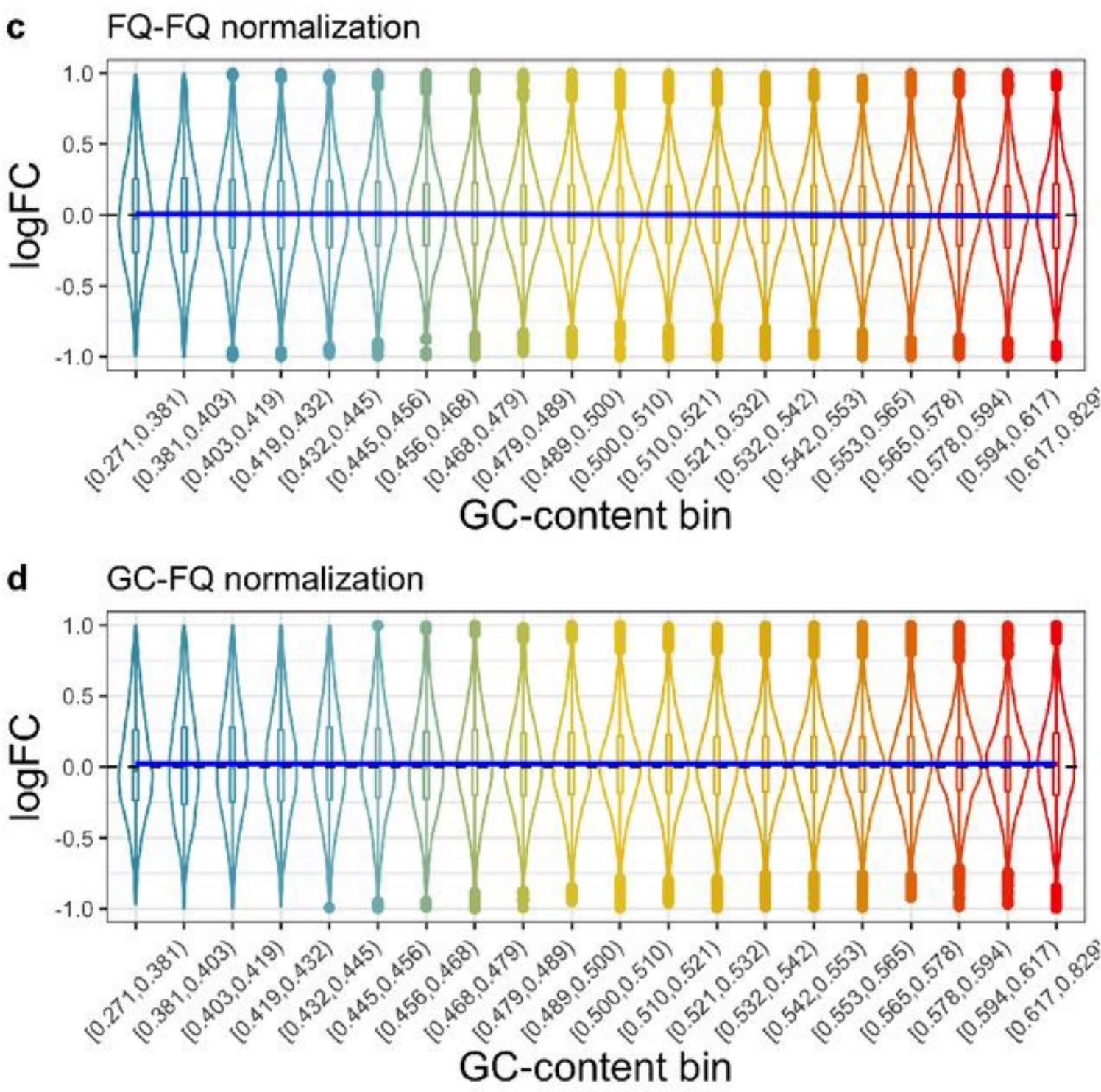
log-count as a function of GC-content



TMM normalisation



GC aware normalisation



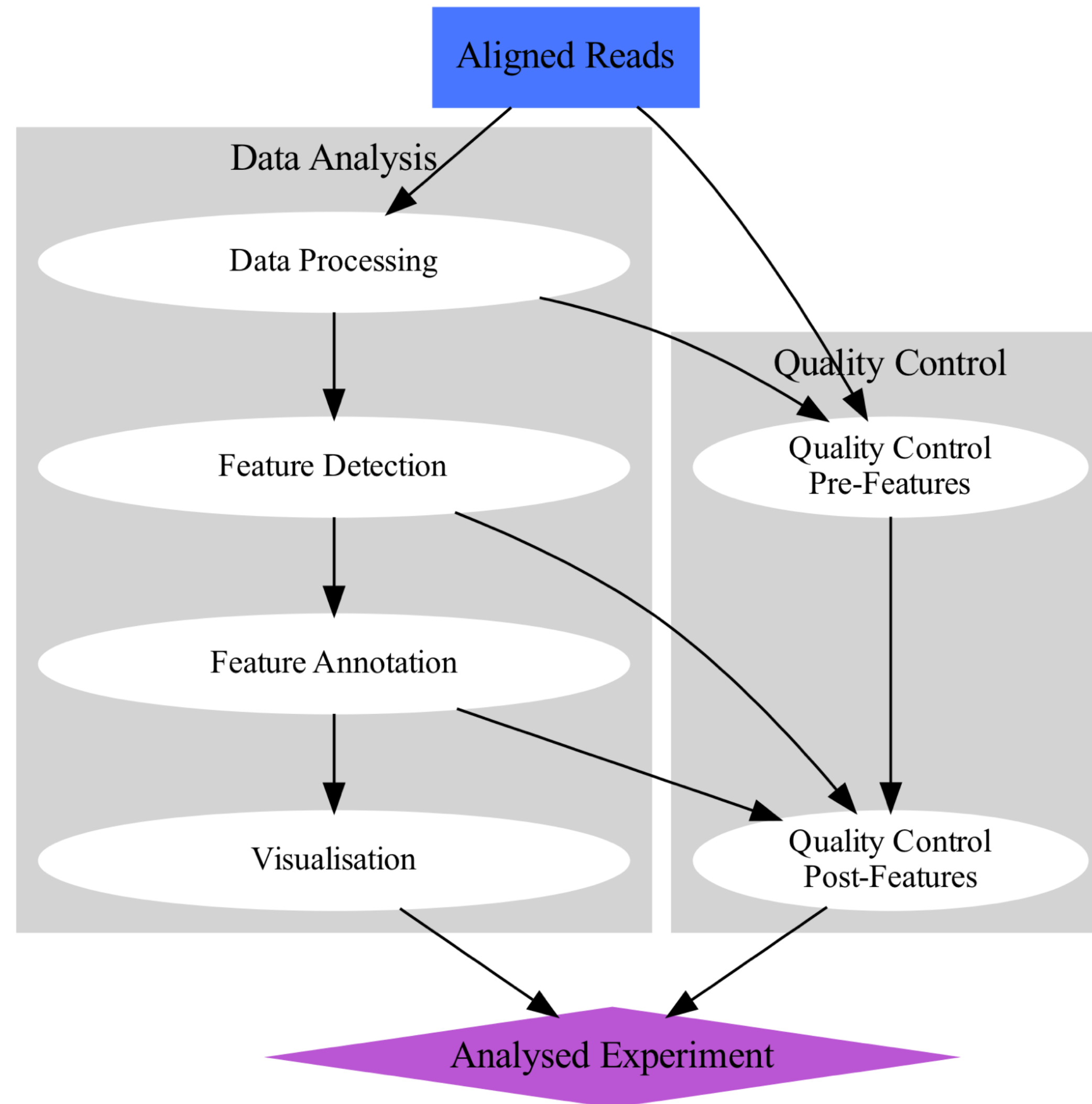
Differential accessibility
log-fold change in bins by GC content

A bias for peaks with low and high GC-content
(in a null setting, LFC should be centred on zero)

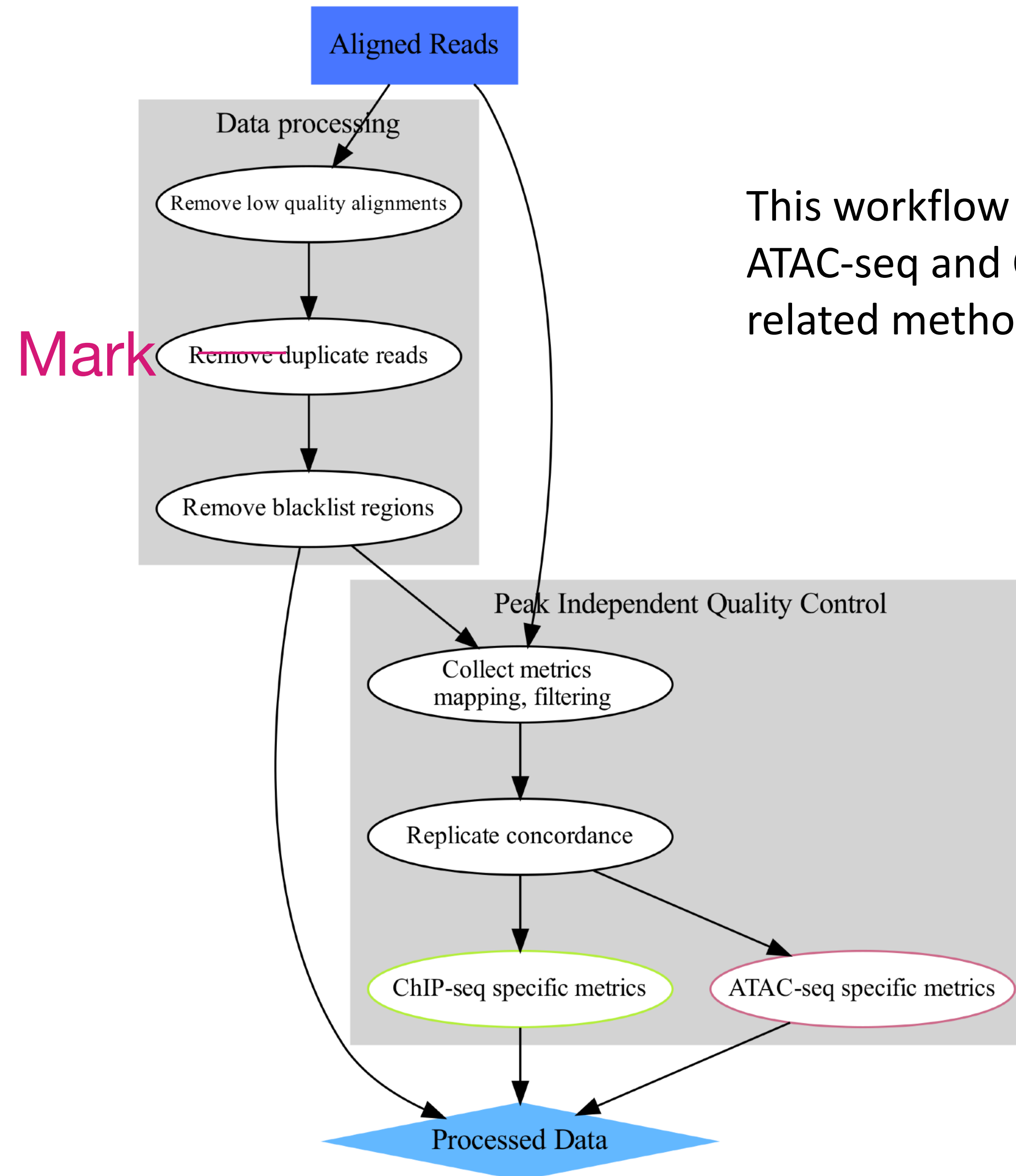
ATAC-seq data analysis

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Exercise Overview

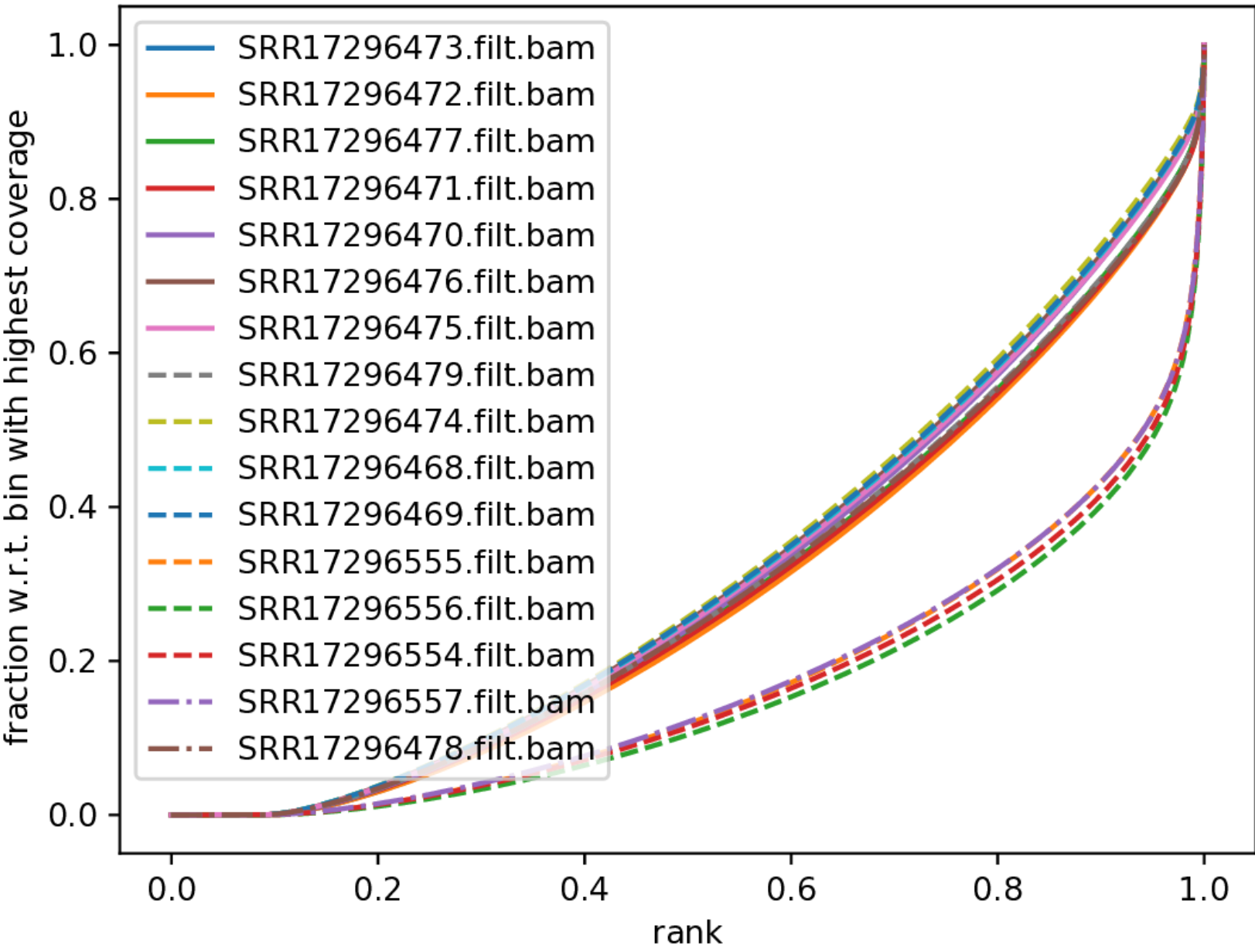


Exercise Overview: Data preprocessing and QC

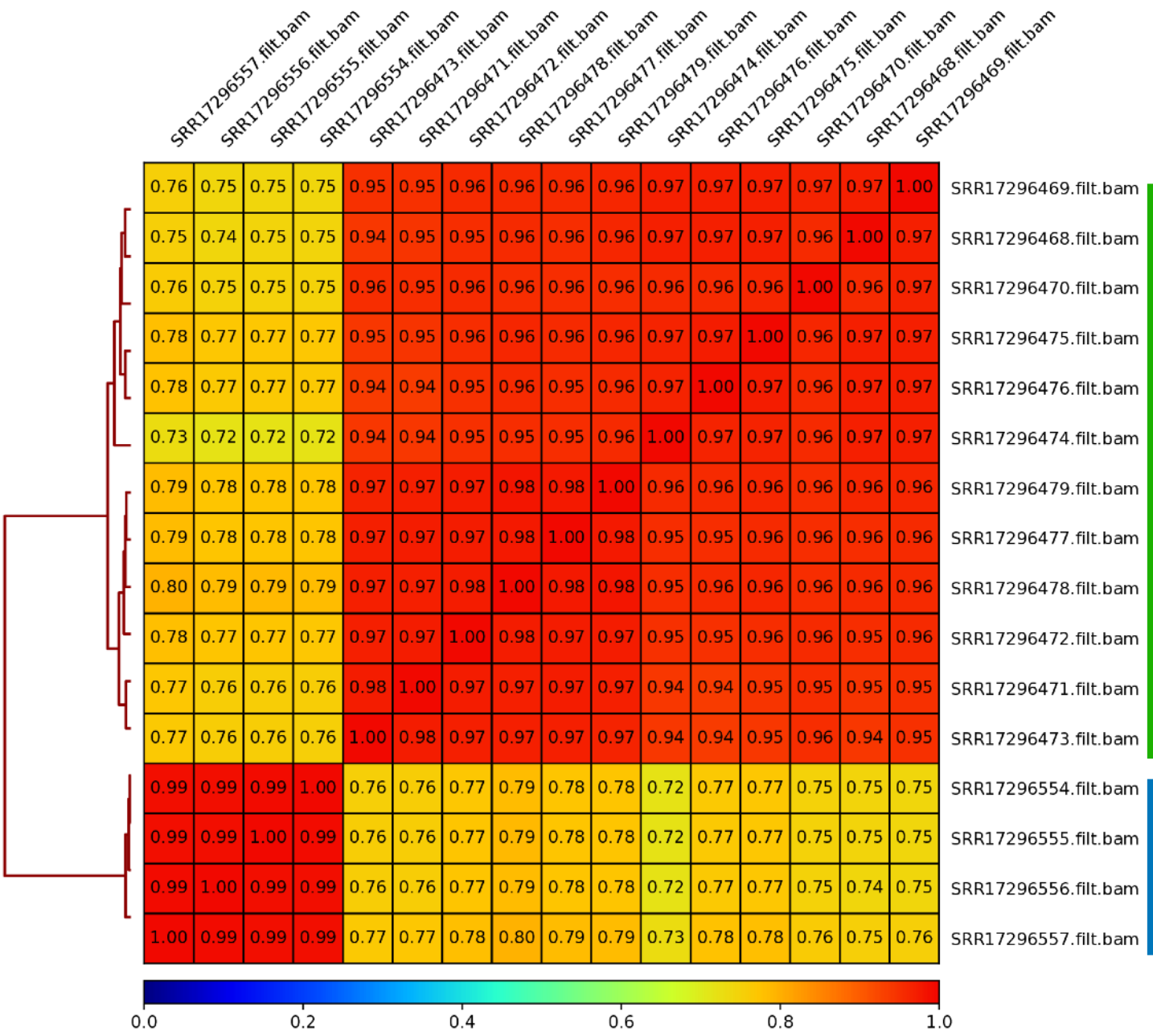


Exercise Overview: Data preprocessing and QC (deepTools)

These QC steps are common for ATAC-seq and ChIP-seq and related methods



Cumulative enrichment ("fingerprint")



Replicate clustering

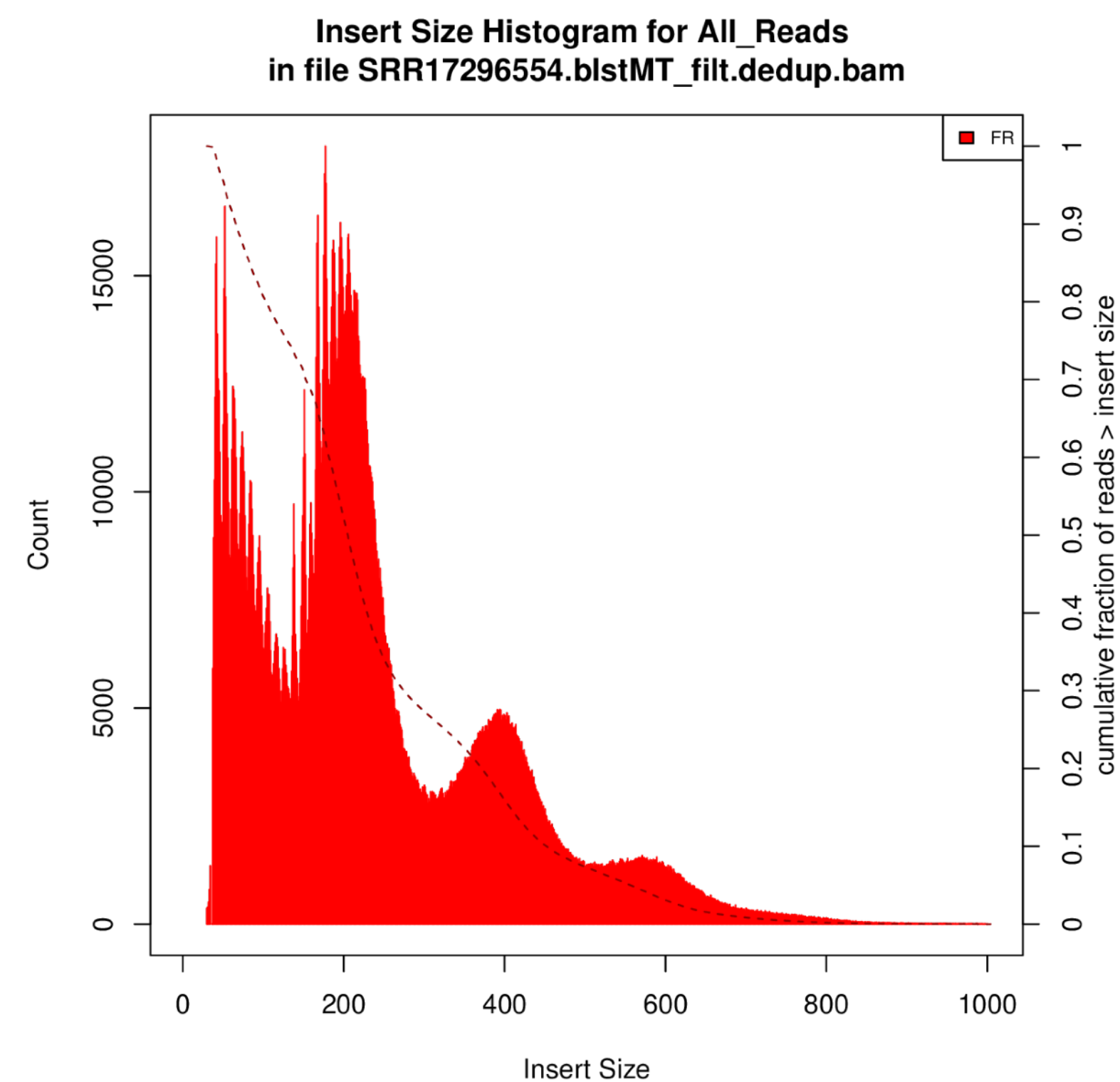
P14 CD8+ T cells
in vitro

P14 CD8+ T cells
in vivo

ctrl

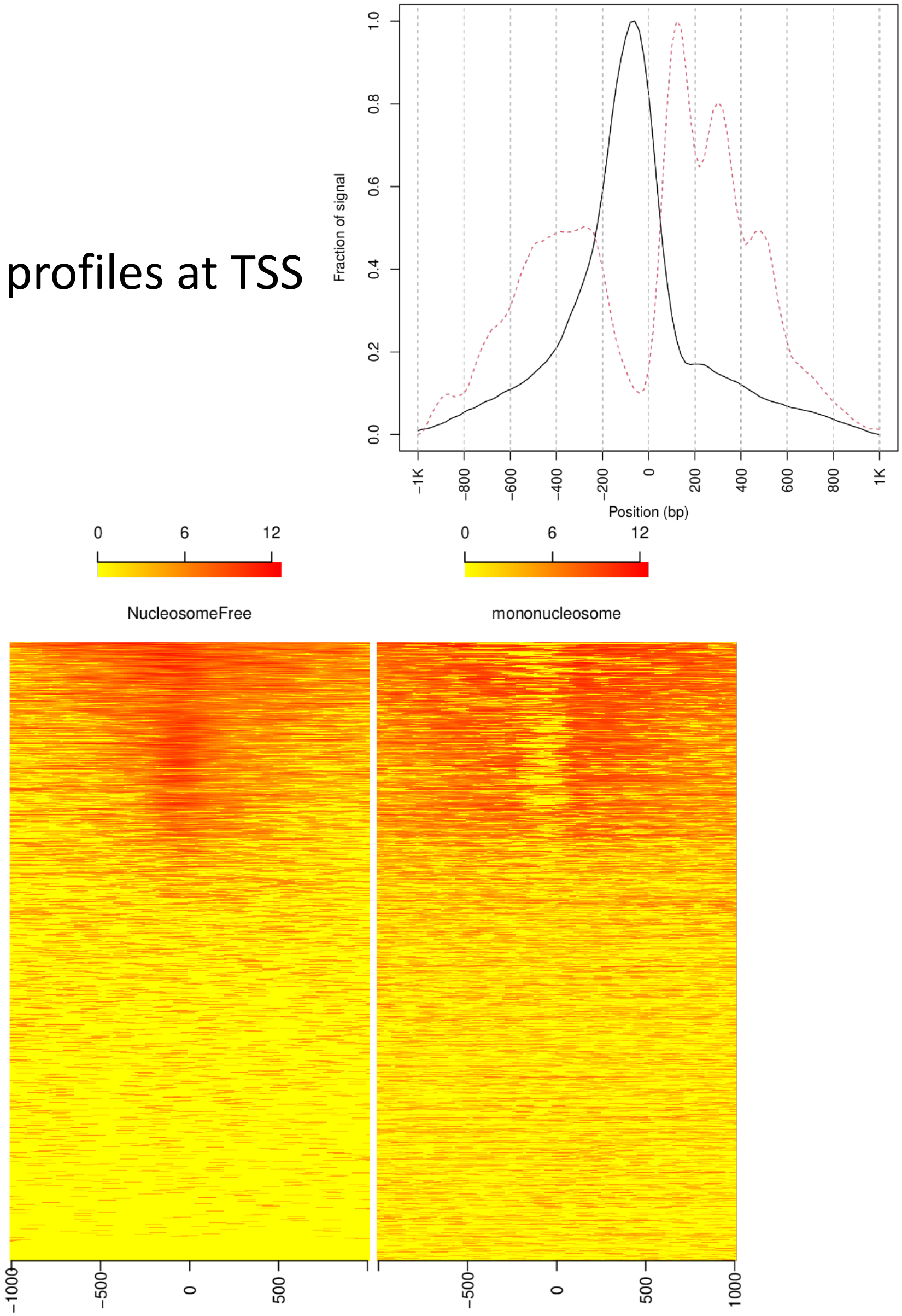
Batf KO

Exercise Overview: ATAC-seq specific QC

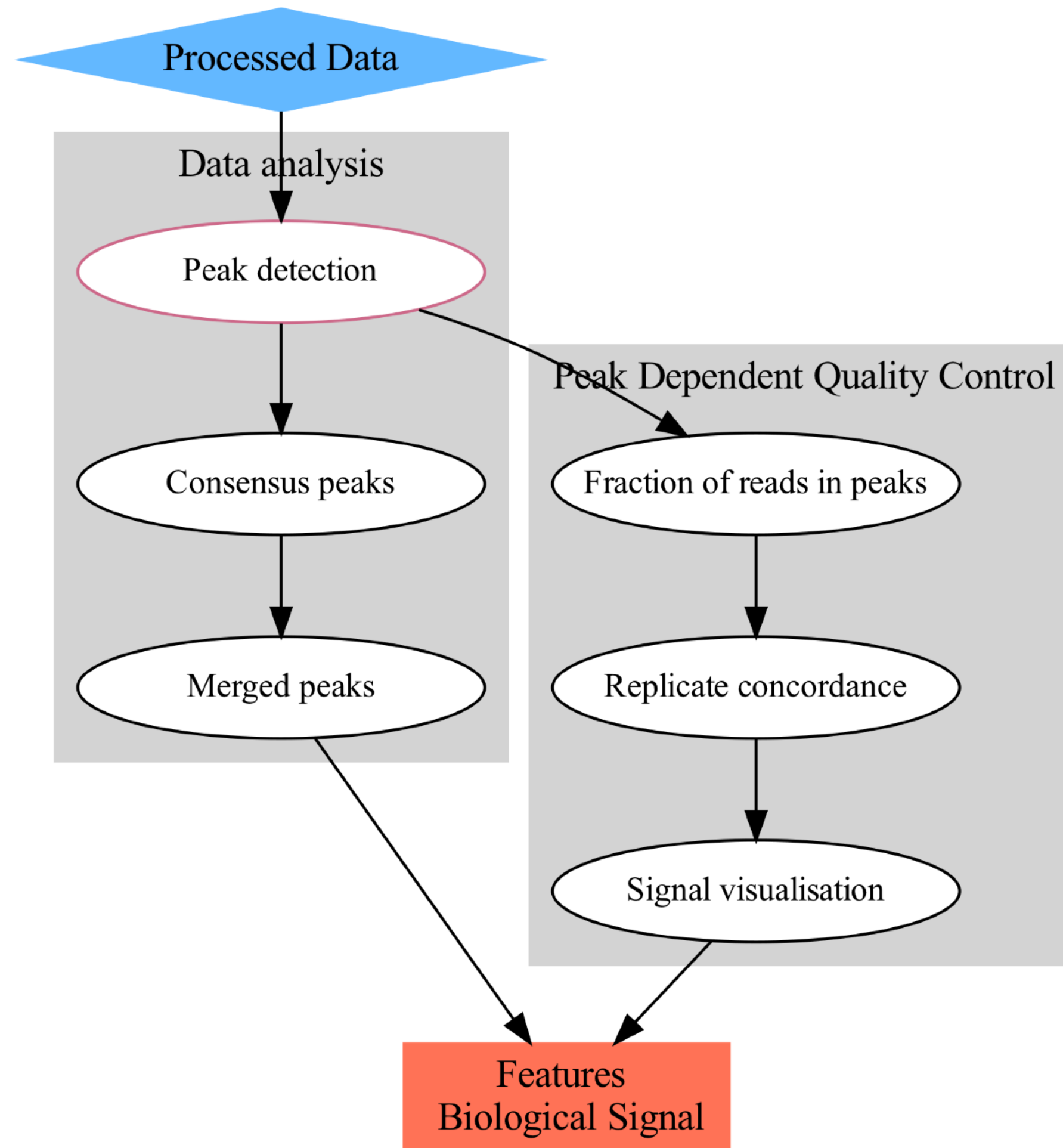


Fragment length distribution

Signal profiles at TSS

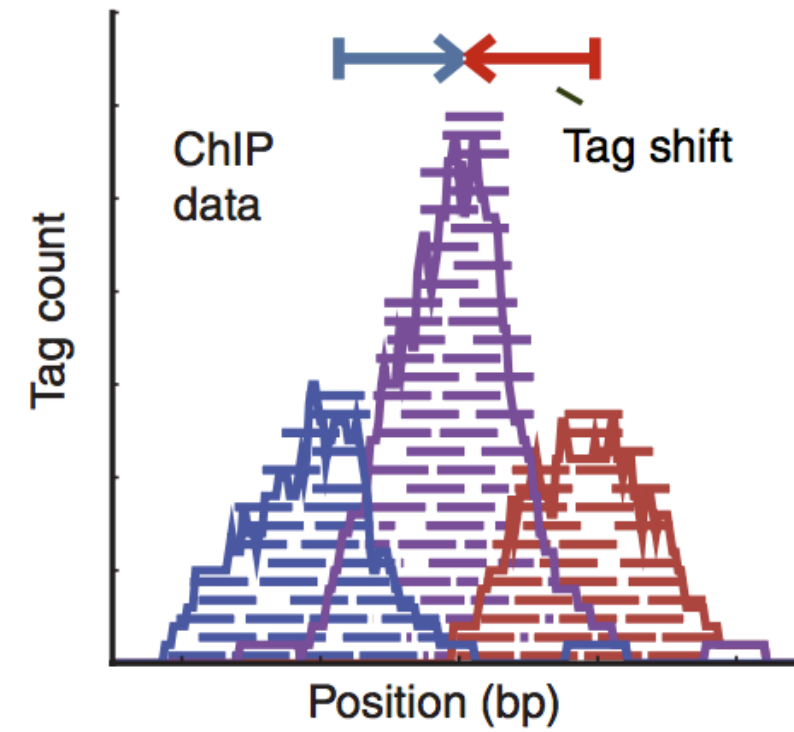


Exercise Overview: ATAC-seq peak calling

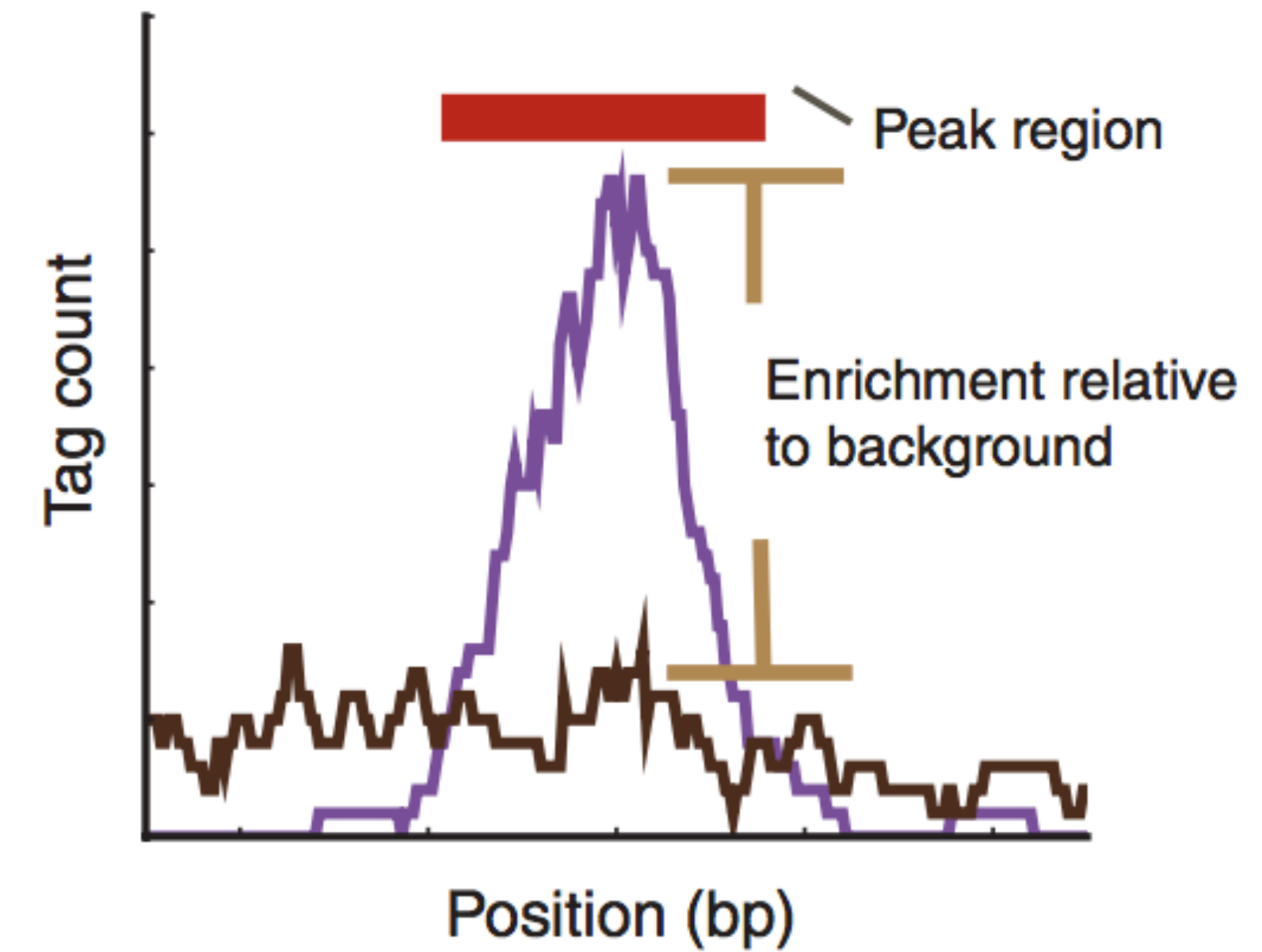
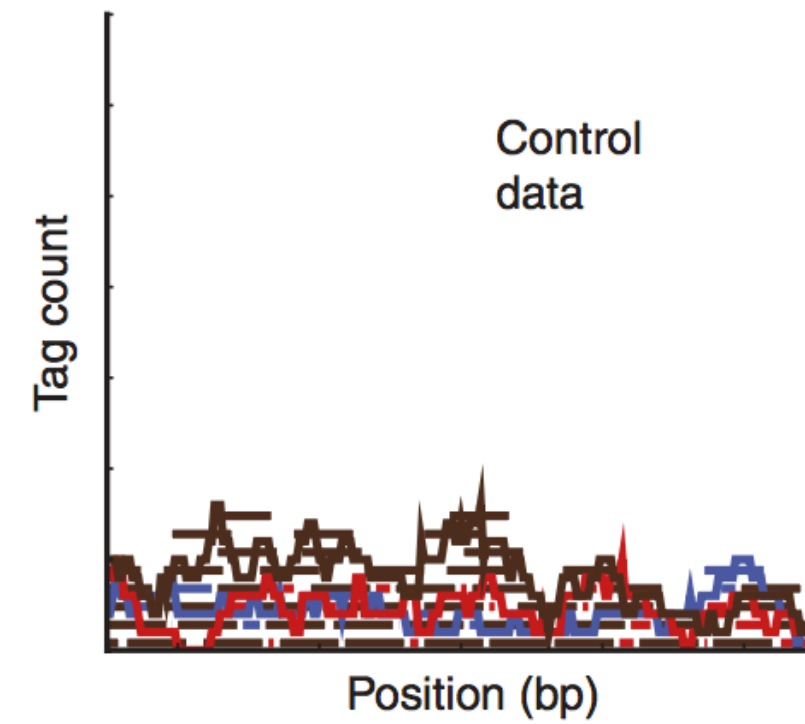


Peak detection

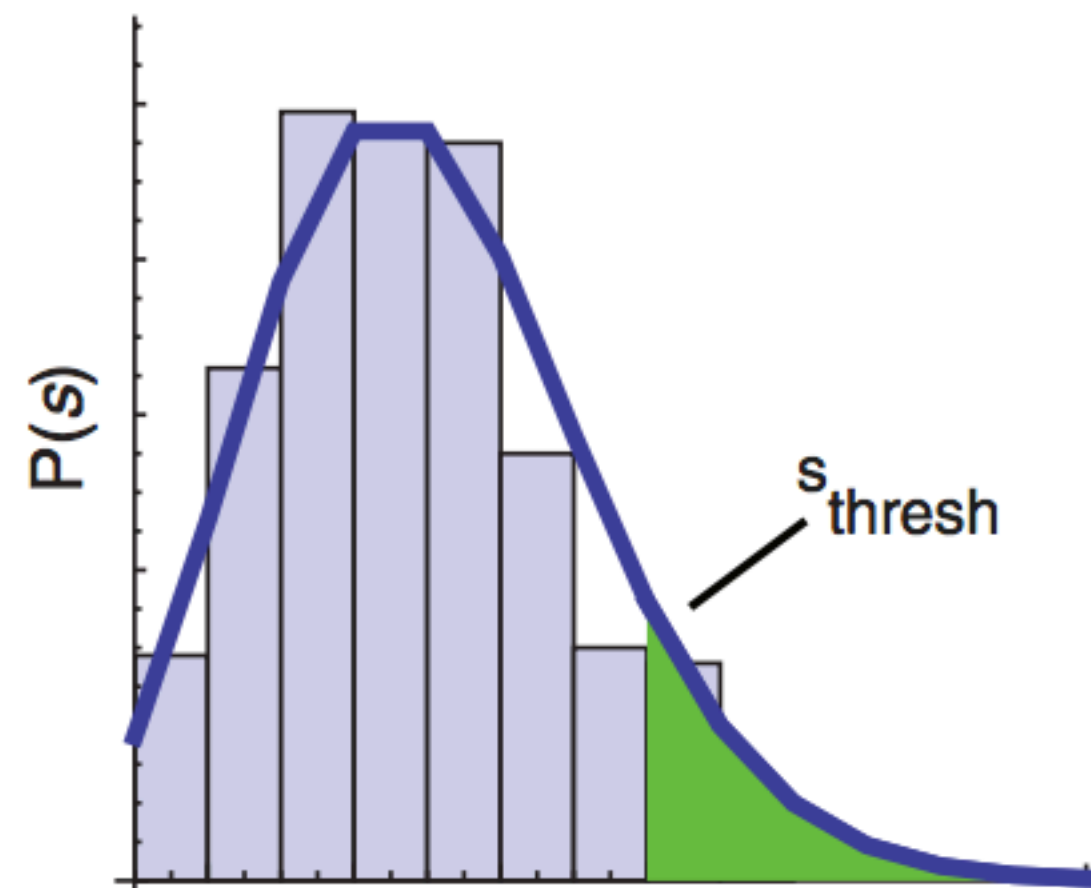
Generate signal profile
along each chromosome



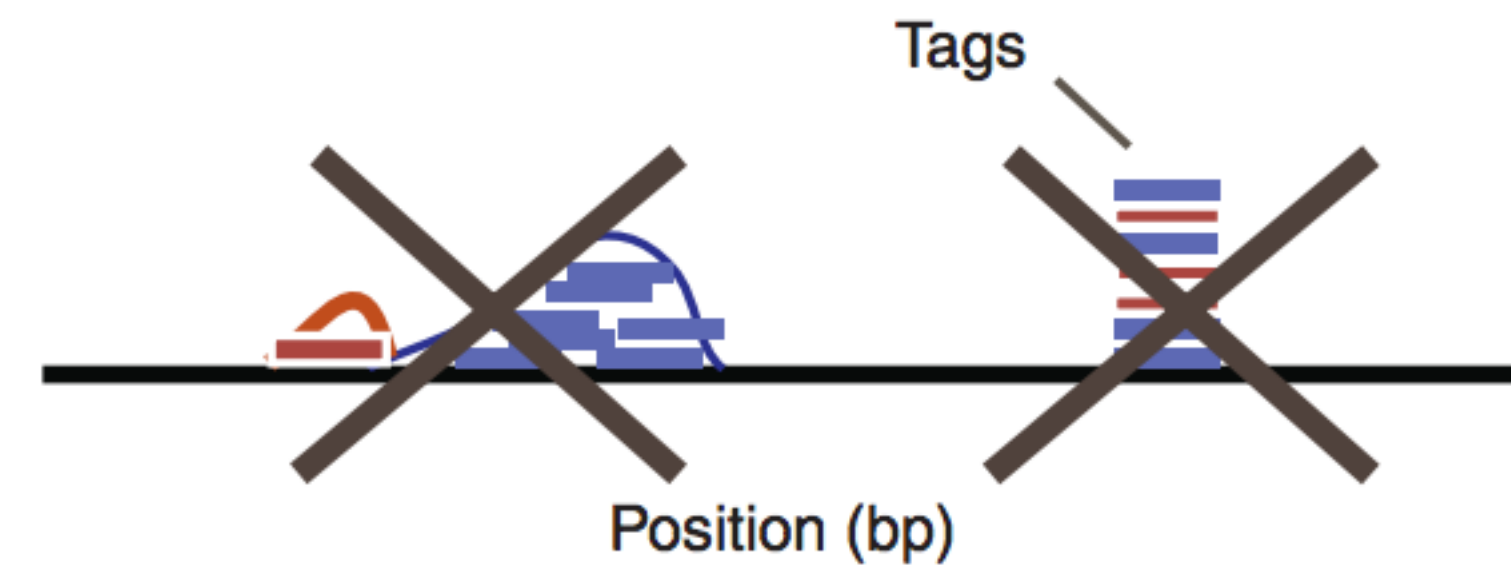
Define background
(model or data)



Assess significance

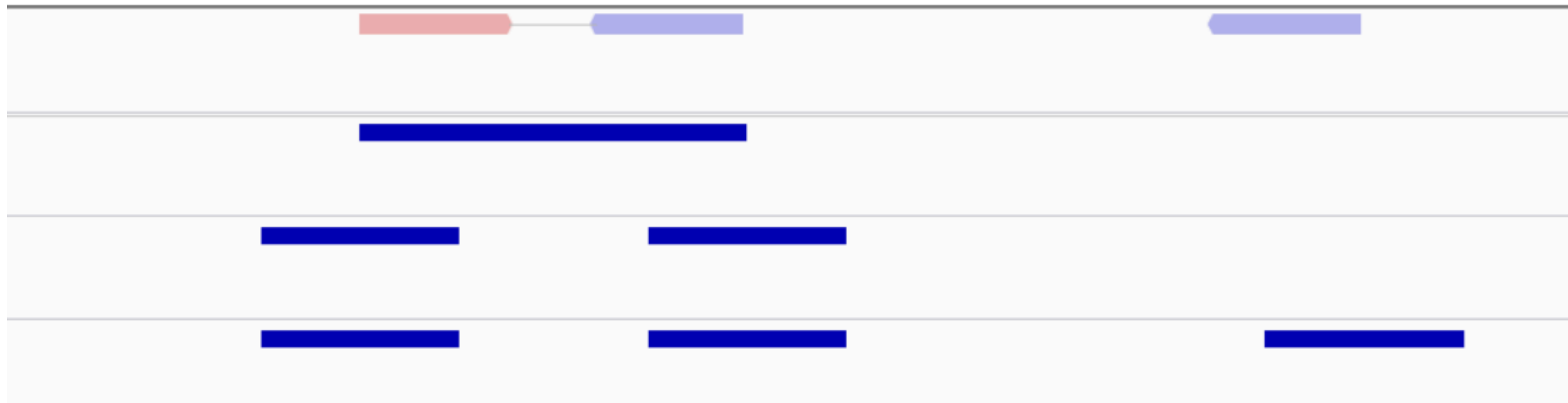


Filter artifacts



Peak detection in ATAC-seq

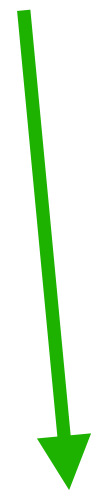
Fragments inferred



default PE (MACS3 broad)

Tn5 cutsites (MACS3 SE narrow, Genrich)

Tn5 cutsites
including unpaired reads

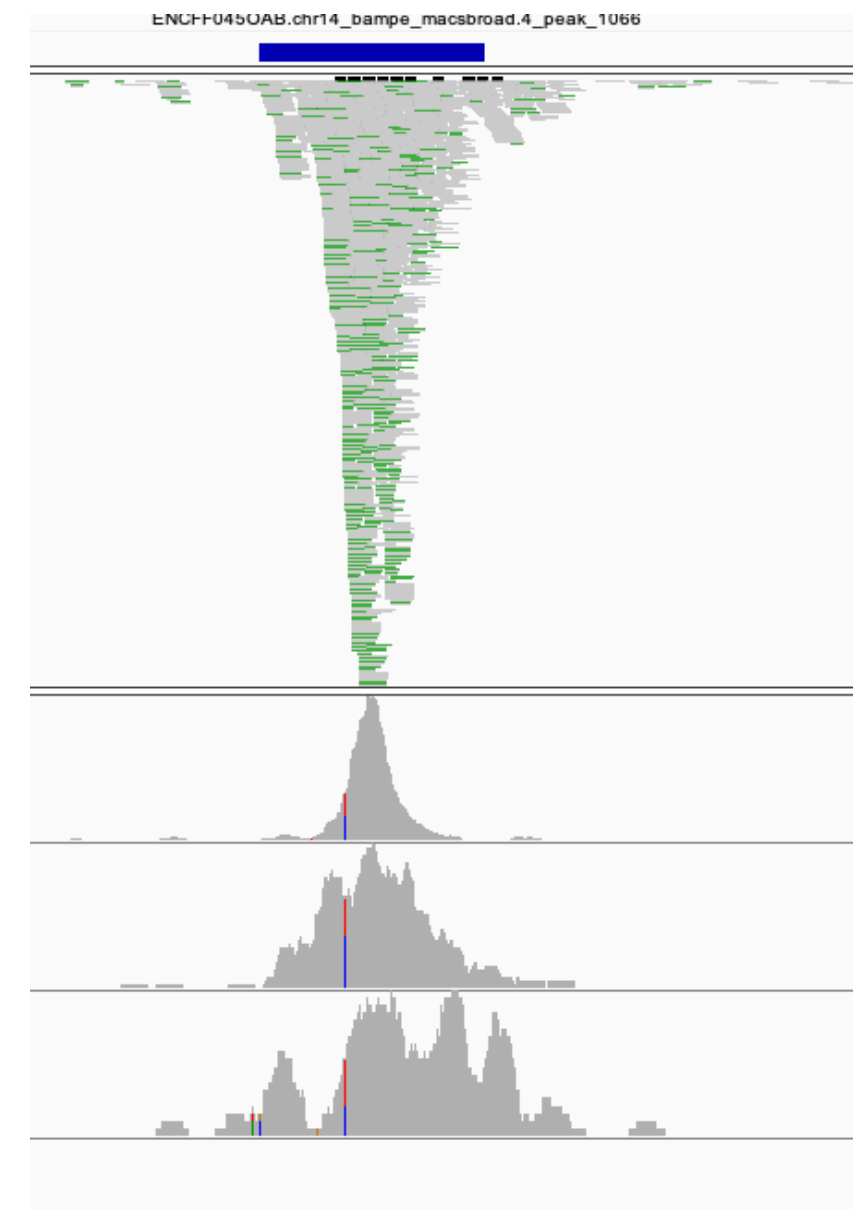


Poisson
distribution

log-normal
distribution

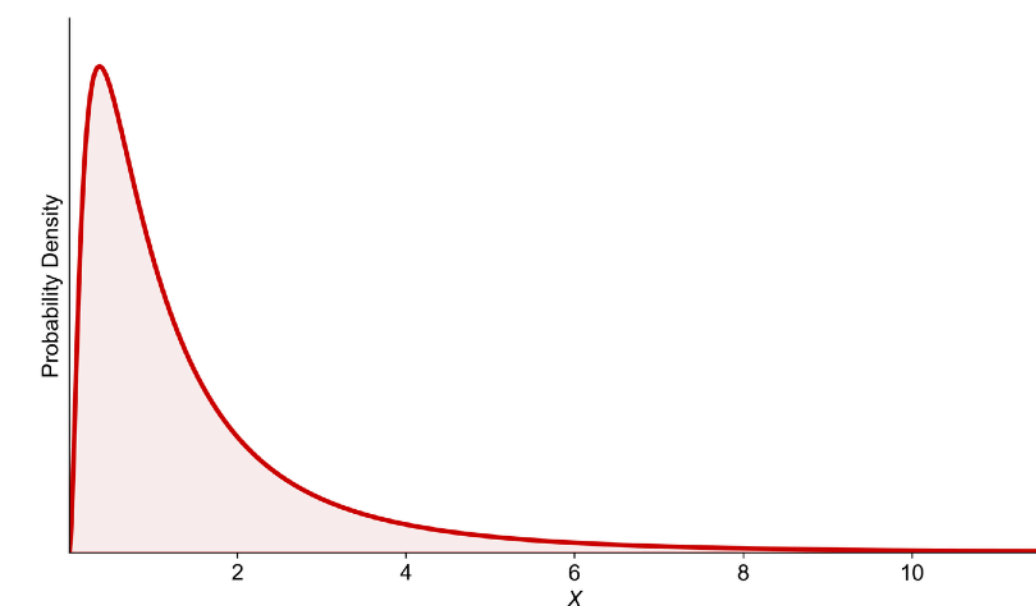
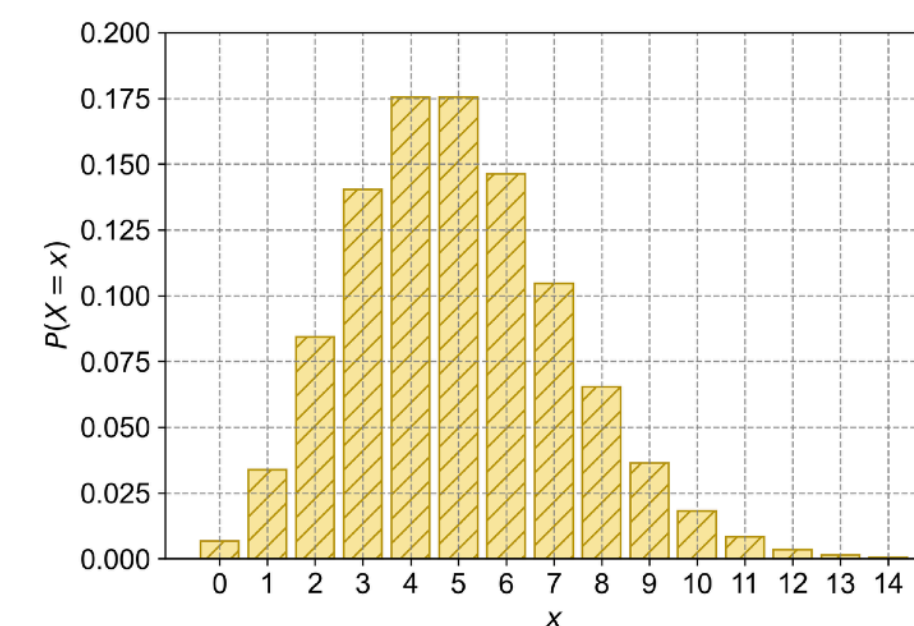
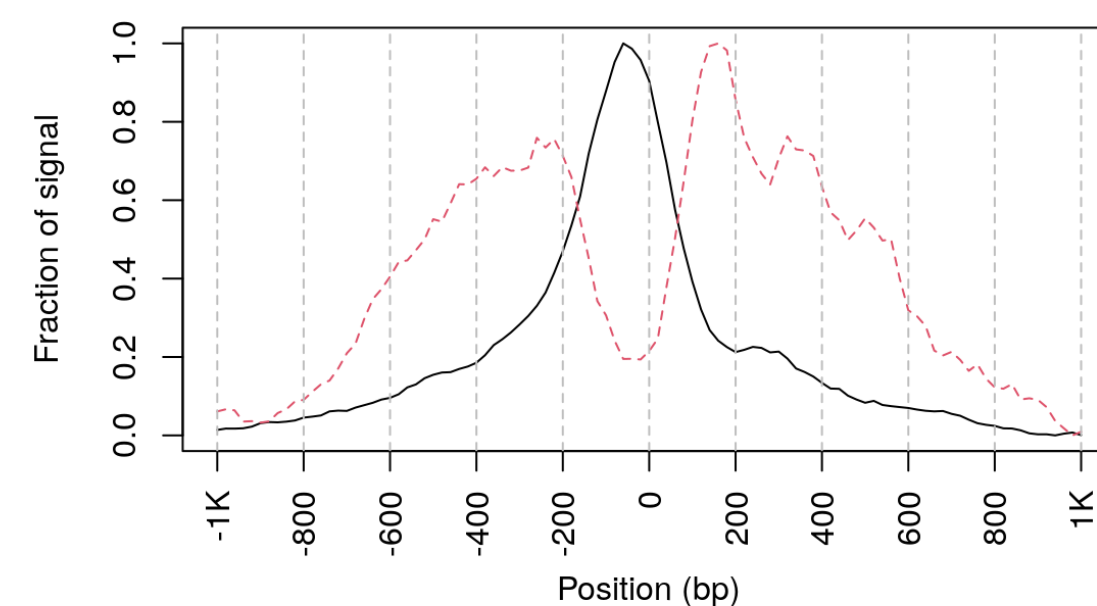
discreet

continuous



HMMRATAC

Analysing signal structure
(NFR peak flanked by mononucleosome peaks)



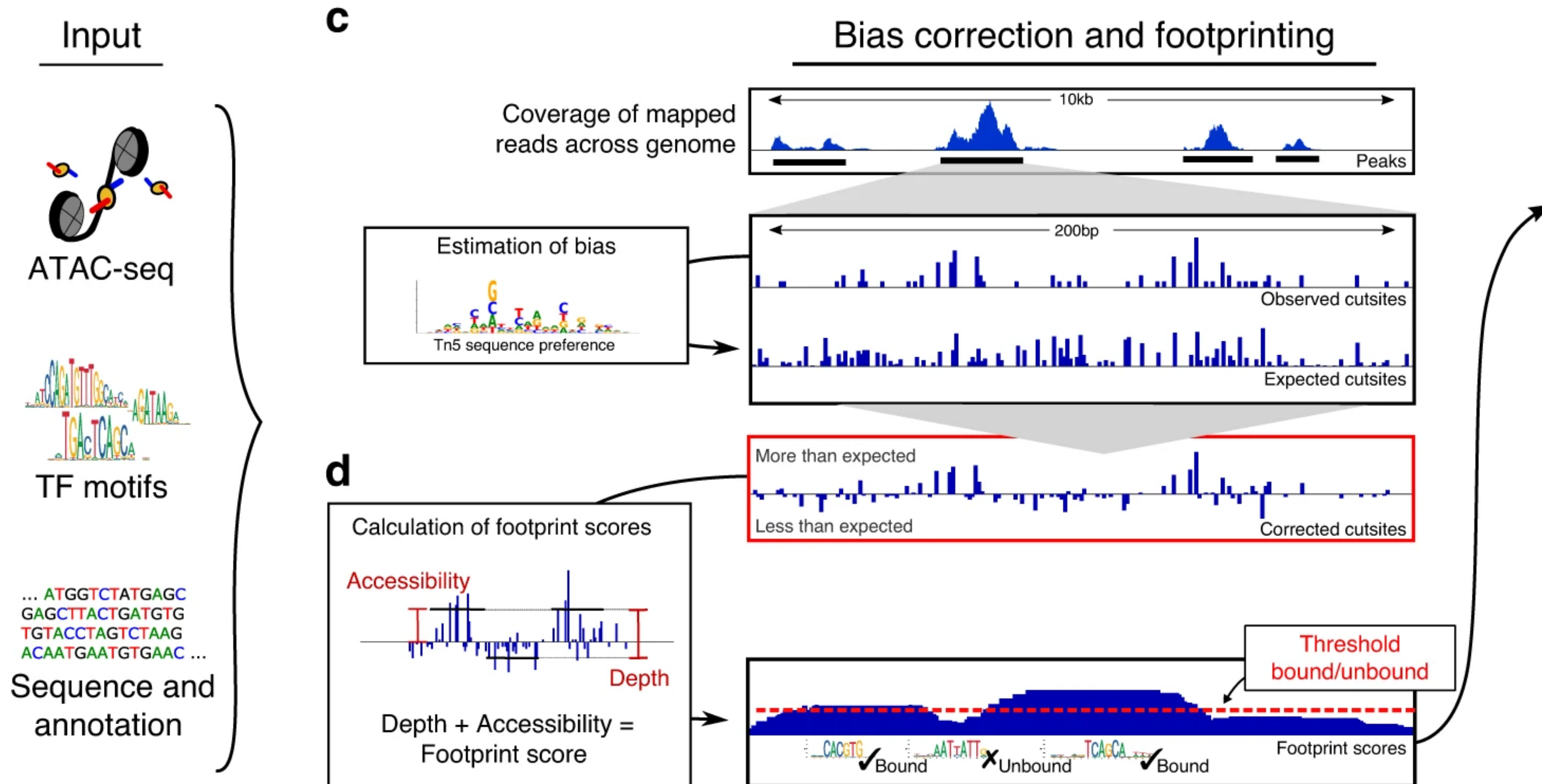
Exercise Overview: ATAC-seq peak calling



- **MACS3 callpeak narrow SE**
- **Genrich**
- **MACS3 callpeak broad PE**

Transcription Factor Footprinting

The TOBIAS footprinting framework



Thank you for listening

Please follow the tutorials:

1. Data processing and QC
2. ATAC-seq peak detection
3. ATAC-seq differential accessibility

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